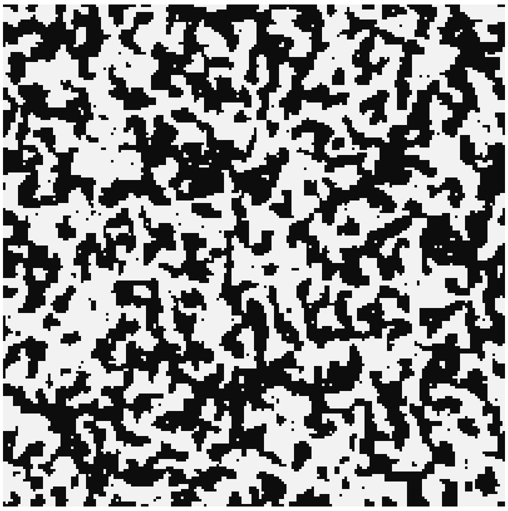
Advanced Statistical Physics

Lecture Notes

Instructor: Prof. Bhavtosh Bansal



Sagnik Seth

Dept of Physical Sciences
IISER Kolkata

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Lecture 01: Introduction and Van der Waal's Equation of State

What we will discuss here in mainly the equilibrium physics of phase transitions which is both interesting and intriguing. Phase transitions are generally the result of the collective behaviour of interacting particles which produce some change in the thermodynamics of the system. For interaction between particles, we require a *pair potential*, which describes the behaviour of the interaction. One of such potentials is the celebrated *Lennard-Jones* potential (LJP) which goes as:

$$u(r) = \varepsilon_0 \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right]$$

where $r = |\mathbf{r}_i - \mathbf{r}_j|$ is the interparticle distance. This is used as a model for intermolecular interactions in many systems. Another "lesser" realistic (but simpler to handle) model can be described by the potential,

$$u(r) = \begin{cases} \infty & r < r_0 \\ -\varepsilon_0 \left(\frac{r_0}{r}\right)^6 & r > r_0 \end{cases}$$

The above potential tells that the minimum possible separation between molecules is r_0, since $u(r)$ blows up below $r = r_0$. Hence the particles act as *hard spheres* (spheres which cannot penetrate each other) of radius $\frac{r_0}{2}$. For $r > r_0$, $u(r) \sim r^{-6}$ which implies *dipolar interactions*.

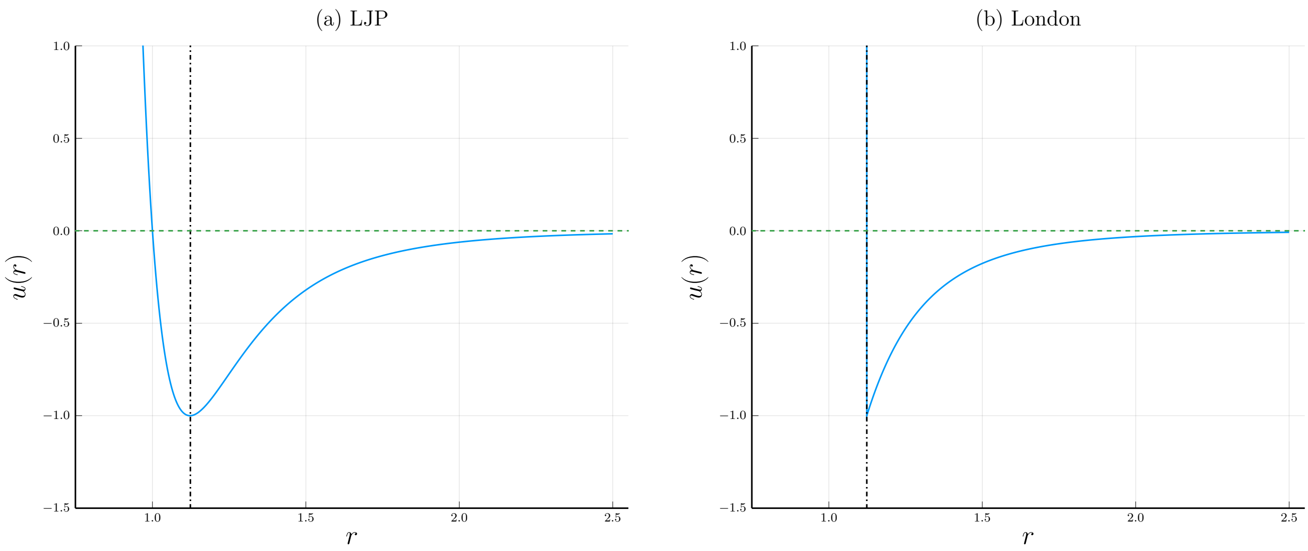


Figure 1: Interaction potentials. (a) Lennard-Jones potential. The vertical line denotes the r at which minima occurs (b) London dispersion potential.

Note that these interactions are mostly *power-law* interactions, $u(r) \sim r^{-s}$. Let d be the dimension of the space in which we are working. Then, it can be shown or motivated that if $s \leq d$, then the interaction is long-ranged which is not ideal for us. Extensivity can break down and some other issues (like negative specific heat) may occur in this case. One such example is that of gravitational systems where $u(r) \sim \frac{1}{r}$. If $s > d$ then the interaction is short-ranged. Since $u(r)$ blows up when $r = 0$, in general, it is advisable to impose a hard core repulsion (like in the London force) which ensures that everything stays okay, since interactions act only after $r > r_0$!

1.1. Van der Waal's gas

Let us see an example of this using the Van der Waal's equation which is a correction to the ideal gas equation $PV = Nk_{\mathrm{B}}T$ and successfully explains the condensation process of gases.

We will work in canonical ensemble here. The *canonical partition function* is written as,

$$\mathcal{Z}(N, V, T) = \frac{1}{N!} \int \prod_i \left(\frac{\mathrm{d}^3 p_i \; \mathrm{d}^3 q_i}{h^3} \right) e^{-\beta \mathcal{H}}$$

where $\mathcal{H}$ is the Hamiltonian of the system. The $N!$ factor has been put to include the indistinguishability of the particles and h^3 factor has been put to make the phase space volume dimensionless. The Hamiltonian consists, as usual, of a kinetic and a pair potential part.

$$\mathcal{H} = \sum_i \frac{p_i^2}{2m} + \frac{1}{2}\sum_{i\neq j} u(|\mathbf{q}_i - \mathbf{q}_j|)$$

The momentum integral can be carried out nicely since this is essentially a Gaussian integral. Using independence of the kinetic terms we get,

$$\mathcal{Z}_{\text{p}} = \left[\int \frac{\mathrm{d}^3 p}{h^3}\, e^{-\beta p^2/2m}\right]^N = \left[\frac{4\pi}{h^3}\int_0^\infty p^2 e^{-\beta p^2/2m}\,\mathrm{d}p\right]^N = \left[\frac{2\pi m k_{\text{B}} T}{h^2}\right]^{3N/2}$$

Defining the thermal wavelength $\lambda_{\text{T}} = {}^{h}/\sqrt{2\pi m k_{\text{B}} T}$, we have

$$\mathcal{Z} = \frac{1}{N!\lambda_{\text{T}}^{3N}}\int \prod_i \mathrm{d}^3 q_i\; e^{-\beta U(\{\mathbf{q}_i\})}$$

where $U(\{\mathbf{q}_i\})$ denotes the interaction term in the Hamiltonian. Tackling this part is extremely hard, since the interactions are between pair of particles, hence independence cannot be assumed.

1.2. Uniform Density Approximation

One of the most popular ways to deal with interactions is through *perturbation*, however, perturbations are generally taken to be *continuous* and hence, non-analyticity (an important aspect of phase transitions) cannot be effectively captured. We will use the method of *mean-field approximation*, which treats the interactions as an "effective potential" which can be found out self-consistently. To simplify our calculations, let us define the number density of the gas, in terms of the delta function,

$$n(\mathbf{q}) = \sum_{i=1}^{N} \delta^{(3)}(\mathbf{q} - \mathbf{q}_i) \tag{1}$$

Then the interaction term becomes,

$$\begin{aligned}
2U &= \sum_i \sum_{j\neq i} u(|\mathbf{q}_i - \mathbf{q}_j|) \\
&= \sum_i \sum_{j\neq i} \int \mathrm{d}q_1\; \delta^{(3)}(\mathbf{q}_1 - \mathbf{q}_i)\cdot u(|\mathbf{q}_1 - \mathbf{q}_j|) \\
&= \sum_i \sum_{j\neq i} \int \mathrm{d}q_1\; \delta^{(3)}(\mathbf{q}_1 - \mathbf{q}_i)\cdot \int \mathrm{d}q_2\; \delta^{(3)}(\mathbf{q}_2 - \mathbf{q}_j)\cdot u(|\mathbf{q}_1 - \mathbf{q}_2|) \\
&= \int \mathrm{d}q_1 \mathrm{d}q_2\; \sum_i \sum_{j\neq i} \delta^{(3)}(\mathbf{q}_1 - \mathbf{q}_i)\cdot\; \delta^{(3)}(\mathbf{q}_2 - \mathbf{q}_j)\cdot u(|\mathbf{q}_1 - \mathbf{q}_2|) \\
&= \int \mathrm{d}q_1 \mathrm{d}q_2\; n(\mathbf{q}_1) n(\mathbf{q}_2) u(|\mathbf{q}_1 - \mathbf{q}_2|)
\end{aligned}$$

We will use a heavily *unmotivated* approximation where we will treat the density to be uniform, independent of the spatial coordinate, that is, $n(\mathbf{r}) \equiv n$ Then,

$$U = \frac{n^2}{2}\int \mathrm{d}q_1^3 \mathrm{d}q_2^3\; u(|\mathbf{q}_1 - \mathbf{q}_2|)$$

Let us define the variables $\mathbf{R} = \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2)$ and $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ from which we have $\mathbf{r}_1 = \mathbf{R} + \frac{\mathbf{r}}{2}$ and $\mathbf{r}_2 = \mathbf{R} - \frac{\mathbf{r}}{2}$. The Jacobian of this transformation can be written as,

$$\mathbf{J} \equiv \frac{\partial(\mathbf{r}_1, \mathbf{r}_2)}{\partial \mathbf{R}, \mathbf{r}} = \begin{pmatrix} \mathbb{1} & \frac{1}{2}\mathbb{1} \\ \mathbb{1} & -\frac{1}{2}\mathbb{1} \end{pmatrix} \implies J = \det(\mathbf{J}) = -1$$

Using the new variables, the integral becomes,

$$U = \frac{n^2}{2}\int \mathrm{d}^3R\ \mathrm{d}^3r\ u(r) = \frac{n^2 V}{2}\int_{r_0}^{\infty} \mathrm{d}^3r\ u(r) = \frac{N^2}{2V}\int \mathrm{d}^3r\ u(r)$$

where we have used $n = {}^{N}\!/_{V}$ and r_0 is the hard-core cutoff. Now, assume a power-law interaction for $r > r_0$, $u(r) = -\varepsilon_0\left(\frac{r_0}{r}\right)^s$ using which we get,

$$U = -\frac{\varepsilon_0 N^2}{2V}\cdot r_0^s \int_{r_0}^{\infty} 4\pi r^2 \mathrm{d}r\ \frac{1}{r^s} = -\frac{4\pi\varepsilon_0 N^2 r_0^s}{2V(3-s)}\ r^{3-s}\Big|_{r_0}^{\infty}$$

As we can see, U is finite only when $3 - s < 0 \implies s > 3$ where 3 was our spatial dimension. Assuming $s > 3$, we have $r^{3-s} \to 0$ and we have

$$U = \frac{4\pi\varepsilon_0 N^2 r_0^s}{2V(3-s)} r_0^{3-s} = -\frac{4\pi\varepsilon_0 N^2 r_0^3}{2V(s-3)} = -\frac{aN^2}{V} \qquad \text{where,}\ \ a \equiv \frac{2\pi\varepsilon_0 r_0^3}{s-3}$$

We get an 'effective' interaction potential using the *uniform density approximation.* From here, the partion function now becomes,

$$\mathcal{Z} = \frac{e^{\beta a N^2/V}}{N!\lambda_{\mathrm{T}}^{3N}}\int \prod \mathrm{d}q_i$$

Now, the integral is simply not equal to V^N, since we have assumed a hard-sphere gas and each particle has some finite volume. Assume that each particle occupies a small volume ω, which cannot be occupied by the other particles. If one particle occupies volume say V, then the second particle can occupy $V - \omega$, the third occupies $V - 2\omega$ and so on. Then the integral will be just,

$$\begin{aligned}
\int \prod \mathrm{d}q_i \equiv \prod_{i=0}^{N-1}(V - i\omega) &\approx \prod (V - j\omega)(V - (N-j)\omega) \\
&= \prod V^2\left(1 - \frac{j\omega}{V}\right)\left(1 - \frac{(N-j)\omega}{V}\right) \\
&\approx V^N \prod \left[1 - \frac{(j + N - j)\omega}{V} + \mathcal{O}(({}^{\omega}\!/_{V})^2)\right] \\
&\approx V^N\left[1 - \frac{N\omega}{V}\right]^{N/2} = \left[V\left(1 - \frac{N\omega}{V}\right)^{1/2}\right]^N
\end{aligned}$$

where we have made the product over $\approx N/2$ pairs. Now, assume $N\omega \ll V$ from which we can use the approximation $(1+x)^n \approx 1 + nx$ to get,

$$\left[V\left(1 - \frac{N\omega}{V}\right)^{1/2}\right]^N \approx \left[V\left(1 - \frac{N\omega}{2V}\right)\right]^N = \left[V - \frac{N\omega}{2}\right]^N$$

We see that the available volume is reduced by a factor of $N\omega/2$. This was the final step and we can now write the partition function,

$$\boxed{\mathcal{Z}(N,V,T) = \frac{e^{\beta a N^2/V}}{N!\lambda_{\mathrm{T}}^{3N}}\left[V - \frac{N\omega}{2}\right]^N} \tag{2}$$

The free energy can be obtained from the partition function using $F = -k_{\mathrm{B}}T\ln\mathcal{Z}$,

$$F = -\cancel{k_{\mathrm{B}}T}\cancel{\beta}\frac{aN^2}{V} - Nk_{\mathrm{B}}T\ln\left[V - \frac{N\omega}{2}\right] + c(N,T)$$

and from here, we can find the pressure $P = -\left(\frac{\partial F}{\partial V}\right)_{N,T}$

$$P = -\frac{aN^2}{V^2} + \frac{Nk_\mathrm{B}T}{\left(V - \frac{N\omega}{2}\right)} \implies \boxed{\left(P + \frac{aN^2}{V^2}\right)(V - Nb) = Nk_\mathrm{B}T} \tag{3}$$

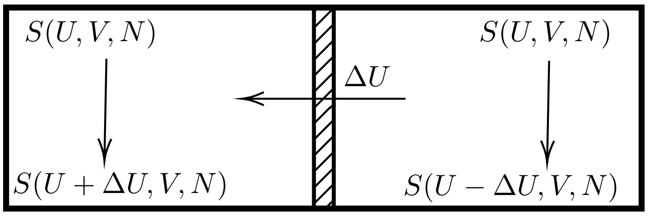
This is the Van der Waal's equation, where we have defined $b = {}^{\omega}/_{2}$. This can be written in an alternate form using the number of moles, $\mathfrak{n} = {}^{N}/_{N_\mathrm{A}}$ where N_A is the Avogadro's number:

$$\left(P + \frac{a'\mathfrak{n}^2}{V^2}\right)(V - \mathfrak{n}b') = \mathfrak{n}RT \iff \left(P + \frac{a'}{v_m^2}\right)(v_m - b') = RT$$

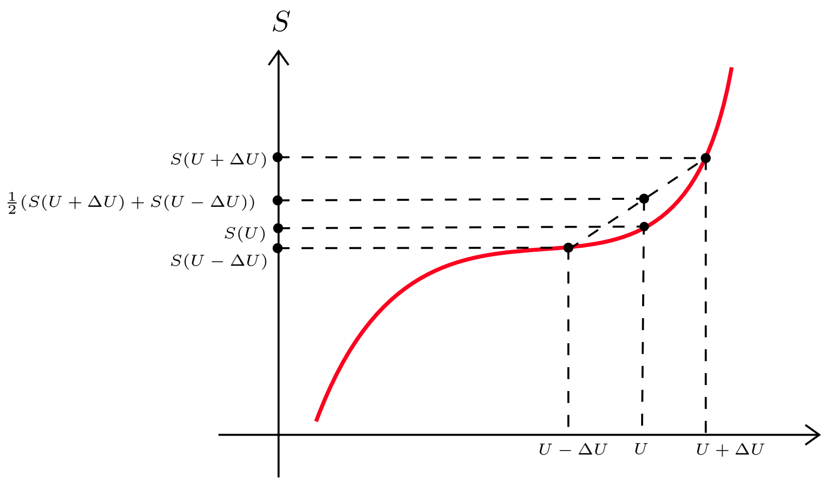
where $a' = N_\mathrm{A}^2 a$ and $b' = bN_\mathrm{A}$ and v_m is the molar volume.

Lecture 02: Stability of Thermodynamic Systems

We will discuss about the conditions under which the system in consideration is stable. This will in turn help us to understand *phase transitions* which are consequences of instability!



Consider two identical subsystems, with entropies $S(U,V,N)$, separated by an adiabatic wall. Suppose we remove an amount of energy ΔU from the first subsystem and transfer it to the second, the total energy changes from $2S(U,V,N)$ to $S(U+\Delta U) + S(U-\Delta U,V,N)$. Suppose that the entropy of the system looked something like in the diagram below.



Then, the resultant entropy after this transfer would be greater than the initial entropy. This would lead to inhomogenities in the system. Within one subsystem, there would be an uneven re-distribution of energy within different regions. This loss of homogenity indicates a *phase separation*!

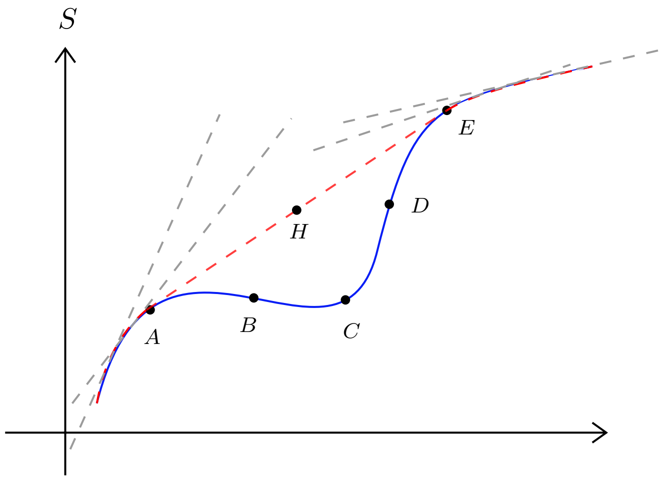
Thus the stability condition is the *concavity of entropy*, that is,

$$S(U-\Delta U,V,N) + S(U+\Delta U,V,N) \leq 2S(U) \tag{4}$$

Taylor expansion of the left hand side leads to the condition (under the limit $\Delta U \to 0$)

$$\boxed{\left(\frac{\partial^2 S}{\partial U^2}\right)_{V,N} \leq 0} \tag{5}$$

This implies that the curvature of the entropy should be downwards! It might happen that we obtain a 'locally convex' entropy curve from some calculation or from experimental data, which does not obey concavity.



This *convex intruder* cannot describe the equilibrium state and we have to obtain the *stable thermodynamic fundamental equation* from this curve. To do that, we construct all the *superior tangents*, that is, tangents which are above this curve. Now, within the family of tangents, there will be one line which will touch the curve at two places.

In the diagram, tangent at any point after A (but before E) will cross the curve. Now, the tangent at A will touch the curve at E. If it had not touched, it would have either intersected the curve or gone above E (If it has intersected the curve, then the tangent at A should not have been considered at all since we are considering only superior tangents. If it had gone over E, this means that there must be some points after A, where this touching will occur. In any case, what we want to say is that, there will be a 'common tangent' at points A and E).

Now, take the envelope of this family of tangents and this will be the fundamental thermodynamic equation that describes the equilibirum states.

Any point on the straight line denotes a phase-separated state, where part of the system is in state A and another part is in E. This happens because of the convex curve of entropy, which creates an inhomogenity.

Note that in the region AB and DE, entropy is still concave, hence these are called *metastable* or locally stable states. These states satisfy Eq.(5) but does not satisfy Eq.(4) while the region BCD satisfy neither condition.

NOTE:

A generic condition for stability is given by $\delta P \delta V \leq 0$ where δP and δV represent a small deviation from the equilibrium value. We can write,

$$0 \leq \delta V \left[\left.\frac{\partial P}{\partial V}\right|_T \delta V + \frac{1}{2}\left.\frac{\partial^2 P}{\partial V^2}\right|_T (\delta V)^2 + \frac{1}{6}\left.\frac{\partial^3 P}{\partial V^3}\right|_T (\delta V)^3 + \dots \right] \tag{6}$$

The first term has $(\delta V)^2$ which is always positive. Hence to make this term negative, we need

$$\left.\frac{\partial P}{\partial V}\right|_T \leq 0$$

Suppose that $\left.\frac{\partial P}{\partial V}\right|_T = 0$. Then the second term has $(\delta V)^3$ and irrespective of what sign $\left.\frac{\partial^2 P}{\partial V^2}\right|_T$ has, δV can always be chosen to make the overall term positive. Hence the only option we have is that

$$\left.\frac{\partial^2 P}{\partial V^2}\right|_T = 0$$

Similar to the first term, we have the third term as,

$$\frac{\partial^3 P}{\partial V^3}\Big|_T \leq 0$$

These define the stability conditions for our thermodynamic system.

2.1. Stability Relations for Potentials

We will now discuss what are the consequence of stability on the thermodynamic potentials, F and G. We will use the fact that for a mechanically stable system, the *specific heat* and the *isothermal compressibility* must be positive for all temperatures. It can be motivated since compressibility and specific heat are related to number and energy fluctuations, σ_{N} and σ_{E} respectively, both of which are positive.

From appendix A, we see that $S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$ from which we get:

$$\left(\frac{\partial^2 F}{\partial T^2}\right)_{V,N} = -\left(\frac{\partial S}{\partial T}\right)_{V,N} = -\frac{1}{T}C_v \leq 0$$

Similarly, we can use $S = -\left(\frac{\partial G}{\partial T}\right)_{P,N}$ from which we get:

$$\left(\frac{\partial^2 G}{\partial T^2}\right)_{P,N} = -\left(\frac{\partial S}{\partial T}\right)_{P,N} = -\frac{1}{T}C_p \leq 0$$

From this, we infer that $G(T,P,N)$ and $F(T,V,N)$ are concave functions of temperature. Apart from temperature, since F depends on V and G depends on P, let us check for pressure and volume too. We note that $V = \left(\frac{\partial G}{\partial P}\right)_{T,N}$ and $P = -\left(\frac{\partial F}{\partial V}\right)_{T,N}$

$$\begin{aligned}\left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N} &= \left(\frac{\partial V}{\partial P}\right)_{T,N} = -V\mathcal{K}_{\mathrm{T}} \leq 0\\ \left(\frac{\partial^2 F}{\partial V^2}\right)_{T,N} &= -\left(\frac{\partial P}{\partial V}\right)_{P,N} = \frac{1}{V\mathcal{K}_{\mathrm{T}}} \geq 0\end{aligned} \tag{7}$$

where we have appropriately used the definitions of isothermal compressibility and heat capacity. Using this, we find that G is a concave function of pressure and F is a convex function of volume. From the above expressions in Eq 7, we can also write

$$\left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N} = -\left\{\left(\frac{\partial^2 F}{\partial V^2}\right)_{T,N}\right\}^{-1} \tag{8}$$

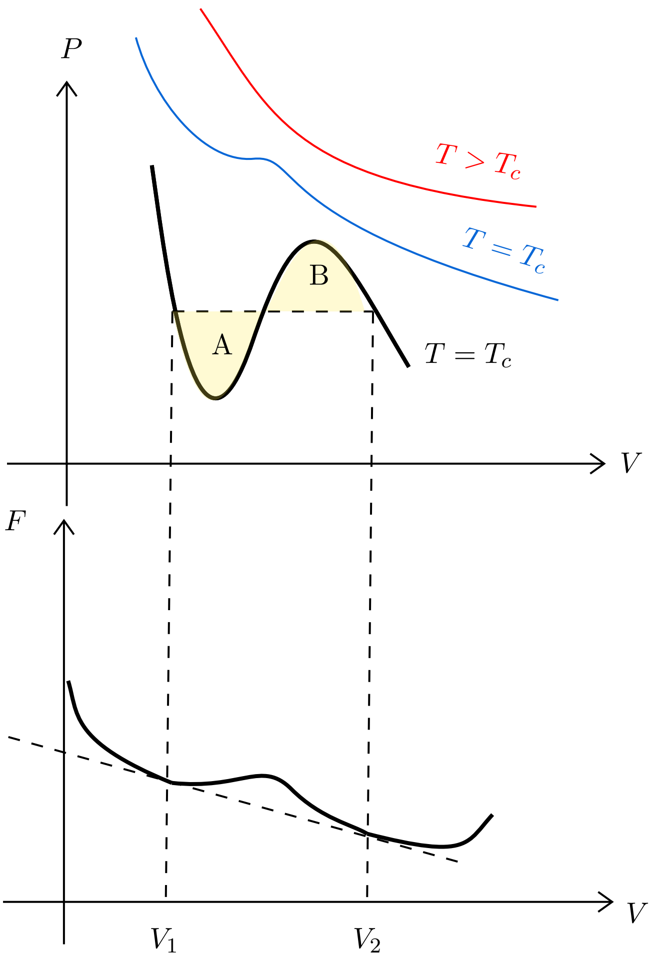
2.2. Stability Analysis of Van der waal's solutions

Recall the Van der Waal's equation from Eq (3),

$$\left(P + \frac{aN^2}{V^2}\right)(V - Nb) = Nk_{\mathrm{B}}T$$

At high temperature, the P vs. V curves at constant temperature (*isotherm*) follows the usual ideal-gas curve. However, there exists a critical temperature T_{c} (which can be found out from the equation), such that, for $T < T_{\mathrm{c}}$, there always exist a region, where the slope $\frac{\partial P}{\partial V}$ is positive. Since $\mathcal{K}_{\mathrm{T}} = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{N,T}$, this implies a negative compressibility, which, as motivated before, is *unphysical.*

This leads to a breakdown of the theory and Maxwell subsequently proposed an *ad hoc* remedy, similar in line to the construction of superior tangents that we noted earlier.



Consider the diagram above. Here, for a subcritical temperature, we see a region between V_1 and V_2 where the slope is positive. This region corresponds to the *concave intruder* (since free energy is a convex function of volume) in the free energy plot. Similarly as before, we replaced the concave part with a straight line denoting the commong tangent. This corresponds to joining the region between V_1 and V_2 in the P-V plot with a straight horizontal line. It can also be shown that the horizontal line is at just such a value of the pressure that the areas labelled A and B are equal and hence this construction is also called *equal-area* construction.

Lecture 03: First Order Phase Transition

Let us now see some qualitative aspect of phase transition. Each phase transition can be viewed as a result of the failure of the stability criteria. We will see some characteristic aspects of first order phase transition[1]. However, let us first note some geometric interpretation of the thermodynamic potentials.

3.1. Geometric Interpretation of Potentials

We will consider the relation $G = F + PV$ which can be obtained from the Legendre transform of the internal energy.

[1]In modern language, these are called *abrupt phase transition*

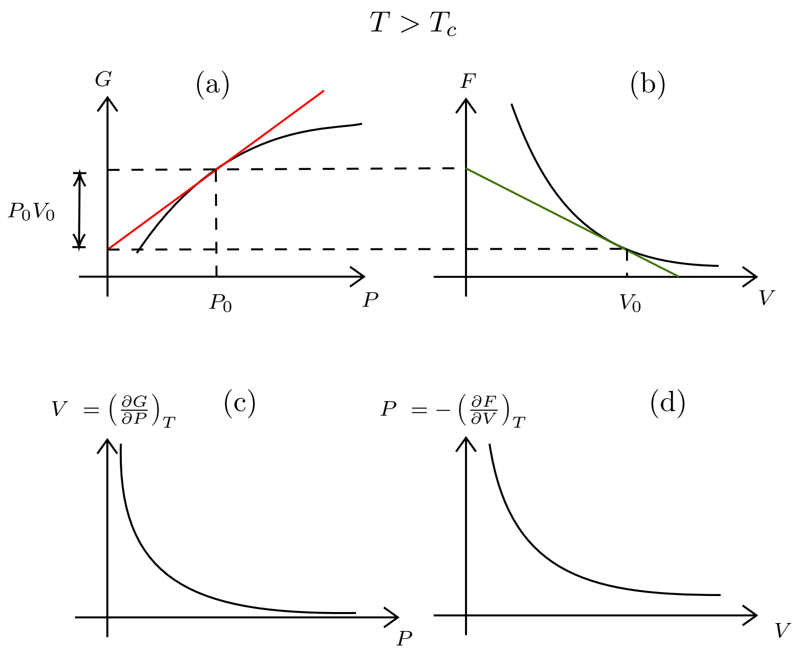


Figure 2: Relation between G and F. (a) Variation of Gibbs potential with pressure at a temperature $T > T_c$ (b) Corresponding plot of Helmholtz potential with V (c) and (d) denotes the P-V diagrams constructed from these potentials

In the above figure, observe the plot of G vs. P, which is smooth. We draw the tangent line at a $P = P_0$ and this intercepts the y-axis somewhere. The equation of the tangent is $G = PV + c$ as $V = \left(\frac{\partial G}{\partial P}\right)_{T,N}$ and it passes through the point (P_0, G_0). Thus we have,

$$G_0 = P_0V_0 + c \implies c = G_0 - P_0V_0$$

The tangent has an intercept of $G_0 - P_0V_0$ and so the remaining length as shown by the arrow in is Fig.2 (a) is $G_0 - (G_0 - P_0V_0) = P_0V_0$. From the length of the intercept of the tangent, we can obtain the Helmholtz potential for this value of P_0. We also obtain the volume from the slope of the tangent. The corresponding F vs. V graph has been plotted using these values. Also, plotted are the V vs. P graph (using the slope of the tangent of the Gibbs potential) and the P vs. V graph (using the slope of the tangent of the Helmholtz potential). This is the case for $T > T_c$. Now let us see the case when $T < T_c$ and the phase transition has occurred. We will see this with respect to the Van der Waal's equation.

$T < T_c$

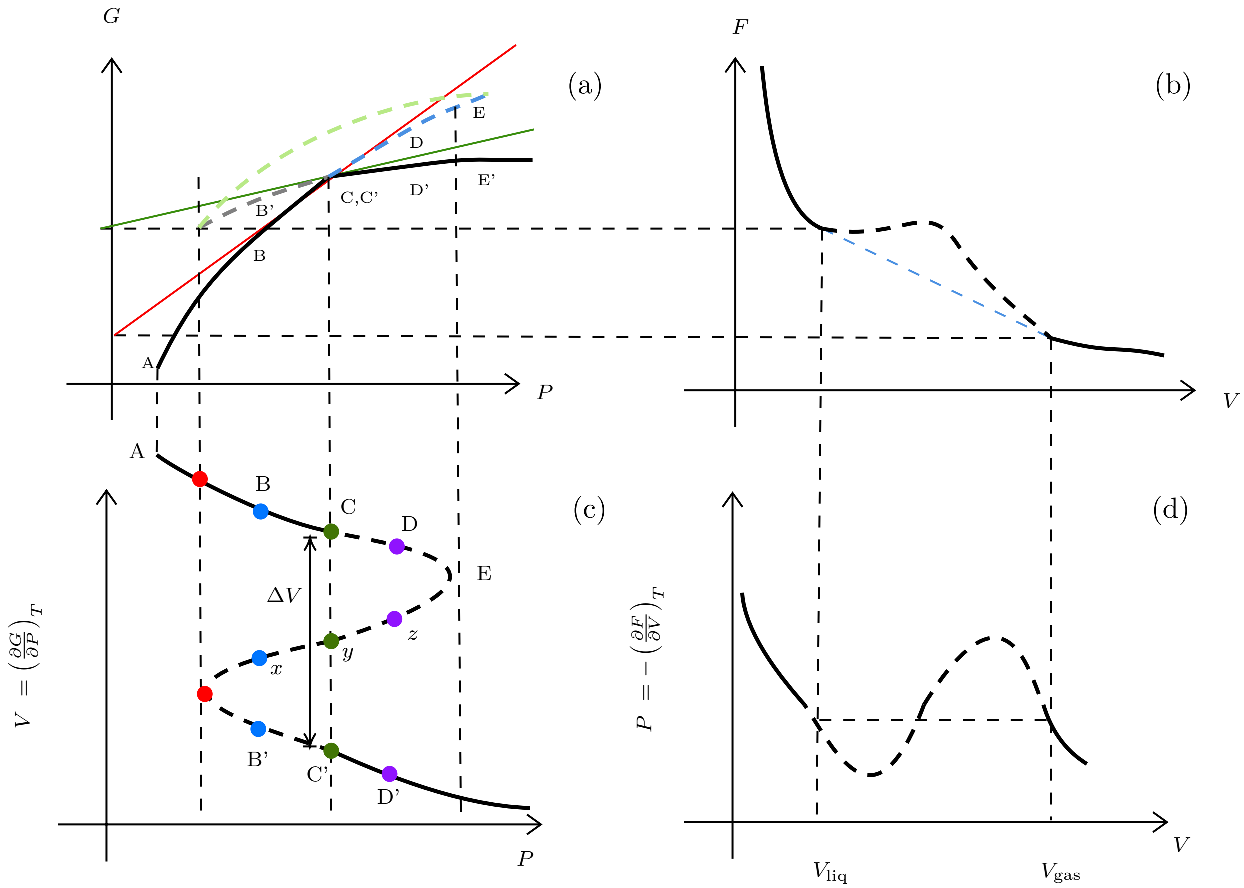


Figure 3: Plot of different quantities for $T < T_c$

In this scenario, G will have a 'kink' for some pressure $P = P_0$, since for first order phase transition, the derivative of the free energy is discontinuous. At this point, $\left(\frac{\partial^2 G}{\partial P^2}\right) \to \infty$ (even though this quantity cannot be defined exactly at $P = P_0$) and using Eq.(8), we get $\left(\frac{\partial A}{\partial V}\right) = 0$. Thus we expect that the corresponding plot of A has a *straight line* portion, as is shown in Fig.3 (b) with a blue dashed line.

Let us now see how the plot of G in (a) is actually constructed. For that, we need the isotherm in (c). Let us start at A and then, integrating we have

$$G(P) - G(P_{\text{A}}) = \int_{P_{\text{A}}}^{P} V \, \mathrm{d}P$$

and we do this for all values of P. This creates a closed loop kind of plot, since for a single P, we have three values of V. Note that, this disappears for higher temperature since then, Van der Waal's equation consists of only one real and two complex roots.

What happens physically? Let us suppose our system is in a temperature bath and the temperature is fixed. So, consider the isotherm in (c) and start increasing pressure from P_A. The system will follow this isotherm. At P_A, there is a unique volume and thus a unique state of the system. Now, if we increase further, for each pressure, we will get multiple values of volume, so there are different states to choose from.

Consider the blue points (all having same P but different V). Of these, the state x is unstable and hence the system has two states B and B' to choose from. The state chosen is the one with the minimum value of G which is determined from panel (a). As clearly seen, the physical state chosen is B. Increasing pressure further, we get to C where the kink is there.

On increasing further, we see that the state chosen is D' since this has lesser value of G. The blue and gray dashed lines in (a) denote the branch of G with higher value of G and thus, is not chosen. The green dashed line represent the unstable region where the states, say x, y or z lies. We see that in (c), no state from the dashed region is selected and hence after C, there is a volume jump, denoted by ΔV. The physical isotherm is thus the one created from the Maxwell equal area construction (as discussed before) and the straight line in the isotherm denotes the *coexistence* of two phases.

The points C and C' are determined by the condition that $G(P_c) = G(P_{c'})$ and hence, $\int\limits_{P_c}^{P_{c'}} V \mathrm{d}P = 0$, which proves that the areas divided by the line segment are indeed equal.

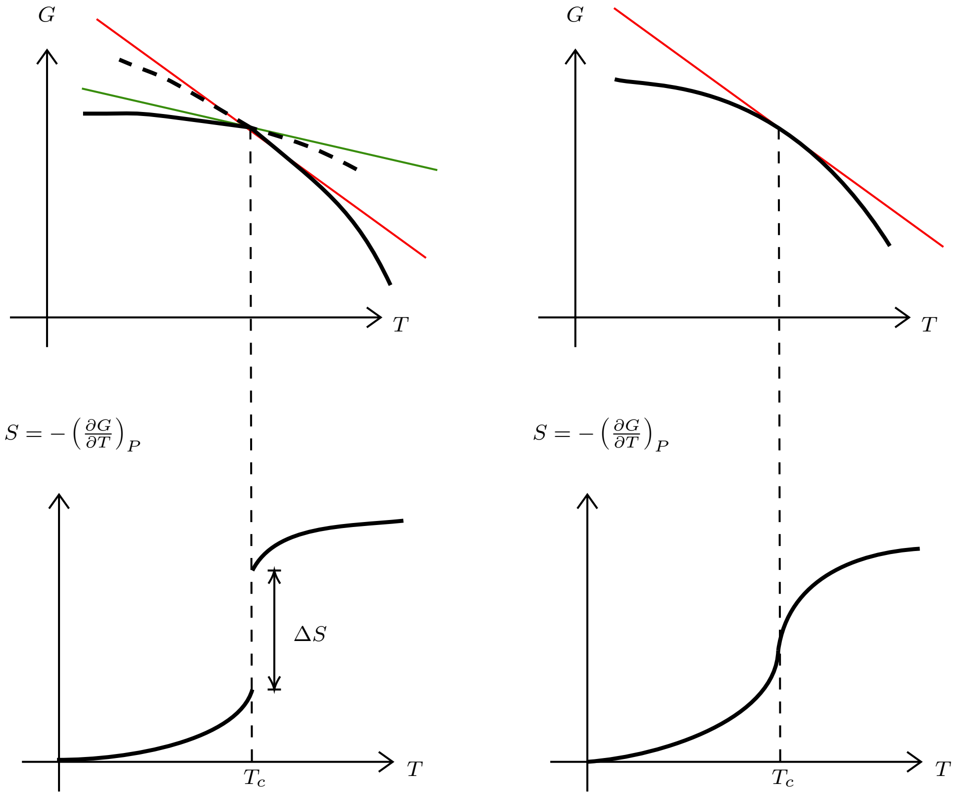


Figure 4: The left column shows the variation of G and S with temperature for first order phase transition, clearly showing an entropy discontinuity. The right column shows this for a higher order phase transition.

There is also an entropy change associated with phase transitions. For first order phase transitions, there is an abrupt entropy jump since entropy is defined as $S = -\left(\frac{\partial G}{\partial T}\right)_P$ and G has a 'kink', making it non-differentiable at the critical point. This entropy jump is associated with the latent heat $L = T_c \Delta S$. For higher order phase transitions, where G is smooth (but some higher order derivative is discontinuous), there is no latent heat since S is continuous. However, the specific heat $C \sim \frac{\mathrm{d}S}{\mathrm{d}T}$ diverges at the critical point.

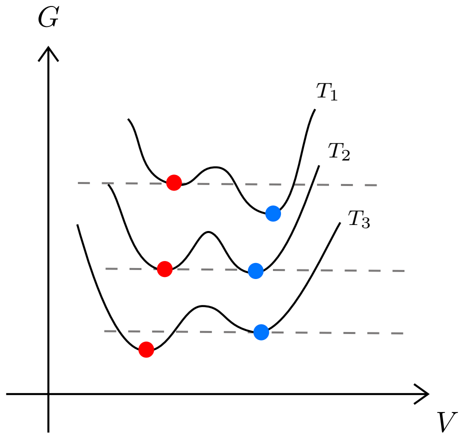


Figure 5: G vs. V plots for different temperatures.

From the previous analysis, we saw that in some interval, we had three values of volume for each pressure (out of which one was unphysical). A physical scenario to imagine is when suppose we have water above its boiling point (so it is in vapour phase). The Gibbs potential has its minima at two volumes as shown in Fig.5 for curve at T_1. The system takes the minima which is lower (the right one).

As we decrease the temperature, at some temperature, both the minima become equal, as for T_2. This is the *critical temperature*. As temperature is reduced further, the other minima becomes the more stable one. The system was in the right minima which was till now the *stable* one. It still remains in the right one, however, it now becomes *metastable*. In a very small timescale, due to some fluctuations, a *nucleus* of condensed liquid (the stable state) forms which grows rapidly and the entire system undergoes the transition.

3.2. Grand canonical basis of phase coexistence

Having seen the physical picture, we will discuss how the liquid-gas phase transition is brought about by considering the partition function that we had found out earlier from Eq(2). However, we used the uniform density approximation for this which is not entirely correct at phase coexistence since the gas (low density) and liquid phases (high density) have weirdly different densities (however, each phase can be assumed to have a uniform density separately).

This problem can be circumvented by considering the *grand canonical ensemble* where we fix the chemical potential μ and the number of particles (and in turn, the density) is appropriately adjusted for different phases. The grand partition function Ξ is defined as,

$$\Xi(\mu, T, V) = \sum_{N=0}^{\infty} e^{\beta\mu N} \mathcal{Z}(N, T, V) \tag{9}$$

where $\mathcal{Z}$ is the canonical partition function. We can appropriately modify the canonical partition function to write is as an expontential:

$$\begin{aligned}
e^{\beta\mu N + \ln \mathcal{Z}} &= \exp\left[\beta\mu N\right] \cdot \exp\left[\frac{\beta a N^2}{V}\right] \cdot \exp\left[N \ln\left(V - \frac{N\omega}{2}\right)\right] \cdot \exp\left[-\ln N! - N \ln \lambda_{\mathrm{T}}^3\right] \\
&= \exp\left[\beta\mu N\right] \cdot \exp\left[\frac{\beta a N^2}{V}\right] \cdot \exp\left[N \ln N + N \ln\left(\frac{V}{N} - \frac{\omega}{2}\right)\right] \cdot \exp\left[-\ln N! - N \ln \lambda_{\mathrm{T}}^3\right] \\
&= \exp\left[\beta\mu N + \frac{\beta a N^2}{V} + N \ln N + N \ln\left(\frac{V}{N} - \frac{\omega}{2}\right) - \ln N! - N \ln \lambda_{\mathrm{T}}^3\right] \\
&= \exp\left[\beta\mu N + \frac{\beta a N^2}{V} + \cancel{N \ln N} + N \ln\left(\frac{V}{N} - \frac{\omega}{2}\right) - \cancel{N \ln N} + N - N \ln \lambda_{\mathrm{T}}^3\right] \\
&= \exp\left[\frac{\beta a N^2}{V} + N \ln\left(\frac{V}{N} - \frac{\omega}{2}\right) + N\Delta\right]
\end{aligned}$$

where we have used Sterling's approximation, $\ln N! \approx N\ln N - N$ and $\Delta := (1+\beta\mu - \ln\lambda_{\mathrm{T}}^3)$. We can now use the saddle-point approximation to evaluate the sum, which gives only a significate contribution for some $N = N^*$.

$$\Xi \approx \exp\left[\frac{\beta a N^{*2}}{V} + N^*\ln\left(\frac{V}{N^*} - \frac{\omega}{2}\right) + N^*\Delta\right] \tag{10}$$

From appendix A, the grand potential is,

$$-PV = \Phi = -k_{\mathrm{B}}T\ln\Xi \implies \beta P = \frac{\ln\Xi}{V} \approx \left[\frac{\beta a N^{*2}}{V^2} + \frac{N^*}{V}\ln\left(\frac{V}{N^*} - \frac{\omega}{2}\right) + \frac{N^*}{V}\Delta\right]$$

We define $n^* = \frac{N^*}{V}$, where n^* is the solution of the equation $\frac{\partial\Psi}{\partial n} = 0$ where $\Psi(n)$ is defined as,

$$\Psi(n) = \beta a n^2 + n\ln\left(\frac{1}{n} - \frac{\omega}{2}\right) + n\Delta$$

Carrying out the differentiation $\frac{\partial\Psi}{\partial n}$, we obtain

$$2\beta a n + \ln\left(\frac{1}{n} - \frac{\omega}{2}\right) + \frac{n}{\frac{1}{n} - \frac{\omega}{2}} \times \left(-\frac{1}{n^2}\right) + \Delta = 0 \implies \boxed{\Delta = \frac{1}{1 - \frac{\omega n}{2}} - 2\beta a n - \ln\left(\frac{1}{n} - \frac{\omega}{2}\right)} \tag{11}$$

Substituting this value of Δ in $\Psi(n)$ we get,

$$\Psi(n^*) = \beta a N^{*2} + \cancel{n^*\ln\left(\frac{1}{N^*} - \frac{\omega}{2}\right)} + \frac{n^*}{1 - \frac{n^*\omega}{2}} - 2\beta a n^{*2} - \cancel{n^*\ln\left(\frac{1}{n^*} - \frac{\omega}{2}\right)} = \frac{1}{\frac{1}{n^*} - \frac{\omega}{2}} - \beta a N^{*2}$$

Comparing this with Van der Waal's equation, this gives us $P = P_{\text{can.}}$ where $P_{\text{can.}}$ is the canonical ensemble pressure. This tells us that the grand canonical and the canonical pressure are identical at particular value of the density n^*. Here we have assumed only a unique n^* which maximises the sum. However, the sum in Eq (10) can be peaked at multiple different values of densities and the appropriate uniform density is chosen for that value which maximises $\psi(n)$.

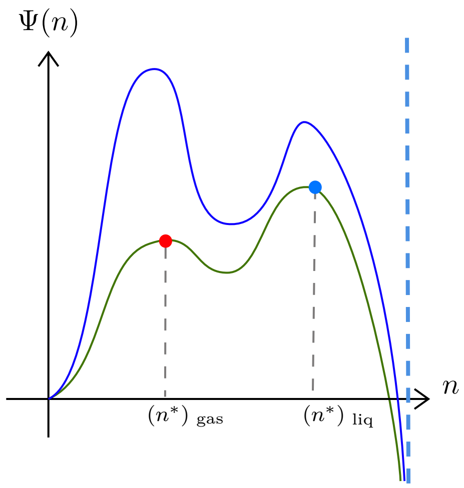


Figure 6: Plot of $\Psi(n)$ vs. n at different temperature. Green plot is for $T < T_c$ and blue is for $T > Tc$. The lower density corresponds to gas and higher density correpsonds to liquid. At $n = \frac{2}{\omega}$, $\Psi(n) \to -\infty$ which has been shown by the blue vertical dashed line.

In the above figure, the blue curve denotes $T > T_c$ and the green curve denotes $T < T_c$. For both $T < T_c$ and $T > T_c$, we have two peaks in the $\Psi(n)$ vs. n plot, however, only the right peak is chosen for $T < T_c$ (corresponding to liquid density) and the left peak is chosen for $T > T_c$ (corresponding to gas density). Considering contribution from both peaks,

$$\ln\Xi \approx \ln\left(e^{V\Psi(n^*_{\text{liq}})} + e^{V\Psi(n^*_{\text{gas}})}\right) = V\Psi(n^*_{\text{liq}}) + \ln\left(1 + e^{V(\Psi(n^*_{\text{gas}}) - \Psi(n^*_{\text{liq}}))}\right)$$

In the thermodynamic limit, if $T > T_c$, $\Psi(n^*_{\text{gas}}) > \Psi(n^*_{\text{liq}})$ then $\ln\Xi \approx V\Psi(n^*_{\text{gas}})$ and hence the gas phase dominates. For $T < T_c$, $\Psi(n^*_{\text{gas}}) < \Psi(n^*_{\text{liq}})$ and hence the liquid phase dominates. From this, we can say that only in the thermodynamic limit, there is a *singularity* in the density at $T = T_c$ (there are no phase transitions in finite systems)!

Lecture 04: Yang-Lee Theory for Phase Transition

We will now discuss about a mathematical basis of condensation based on Yang and Lee's theory of phase transition [1] which is based on the analysis of the zeroes of the grand partition function. The partition function seems to be an analytic function since it essentially consists of sums of well-behaved exponentials.

However, phase transitions are characterised by non-analyticity which leads us to think about how the analytic partition function will capture the essence of the non-analytic phase transition. As discussed before, *true* phase transitions occur only in the thermodynamic limit and in the thermodynamic limit, partition function can indeed become non-analytic (since the limit functions of a sequence of analytic function need not be analytic).

4.1. Singularities and Phase Transition

We will write Eq (9) in terms of the *fugacity* $z = e^{\beta\mu}$ and also consider a finite system with volume V (and then take ultimately take $V \to \infty$). Only a finite number of particles, say M, can be crammed into this finite volume, hence

$$\Xi(z, V) = \sum_{N=0}^{M} z^N \mathcal{Z}(N, V, T) = 1 + z\mathcal{Z}_1 + z^2\mathcal{Z}_2 + \ldots + z^M \mathcal{Z}_M \tag{12}$$

Note that the above equation is a well-behaved polynomial of degree M in z and its coefficients $\mathcal{Z}_N$ are all positive which implies that $\Xi(z, V)$ has no real positive root. From the grand potential $\Phi = -PV = -k_{\mathrm{B}}T \ln \Xi$, we can write the pressure P and density ρ in the thermodynamic limit,

$$\begin{aligned} P &= \lim_{V\to\infty} \frac{k_{\mathrm{B}}T}{V} \ln \Xi(z, V) \\ \langle N \rangle &= z\frac{\partial \ln \Xi}{\partial z} \implies \rho = \lim_{V\to\infty} \frac{\langle N \rangle}{V} = \lim_{V\to\infty} \frac{z}{V}\frac{\partial \ln \Xi(z, V)}{\partial z} \end{aligned} \tag{13}$$

These quantities will not have any singularity unless $\Xi(z, V) = 0$ which cannot hold true, unless we cosider a *complex* fugacity. Then for finite V, the algebraic equation $\Xi = 0$ has M roots appearing in complex-conjugate pairs. As V increases, the number of roots also increases (linearly with V) and moves around in the complex plane, excluding the real axis. In the limit $V \to \infty$, some of these roots touch the real axis and this describes a phase transition. With regard to this, we discuss two theorems as stated in the original paper.

Theorem 1:
For all positive values of z, the limit

$$L = \lim_{V\to\infty} \frac{1}{V} \ln \Xi(z, V)$$

exists and is independent of the shape of the volume holding the particles, with the assumption that the surface area does not scale faster than $V^{2/3}$.

Moreover, $L(z)$ is a continuous, monotonically increasing function of z.

Theorem 2:
Suppose in the complex z plane, there exists a region $\mathcal{R}$ containing the positive real axis, which has no roots of Ξ, then $\lim_{V\to\infty} \frac{1}{V}\frac{\partial^k \ln \Xi}{\partial z^k}$ is uniformly convergent and the limit function is analytic for all $z \in \mathcal{R}$.

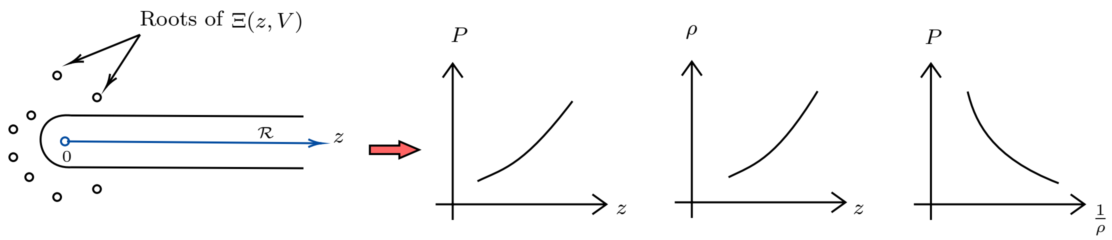
Moreover, the operations $\lim_{V\to\infty}$ and $z\frac{\partial}{\partial z}$ can be interchanged.

The above theorem implies that,

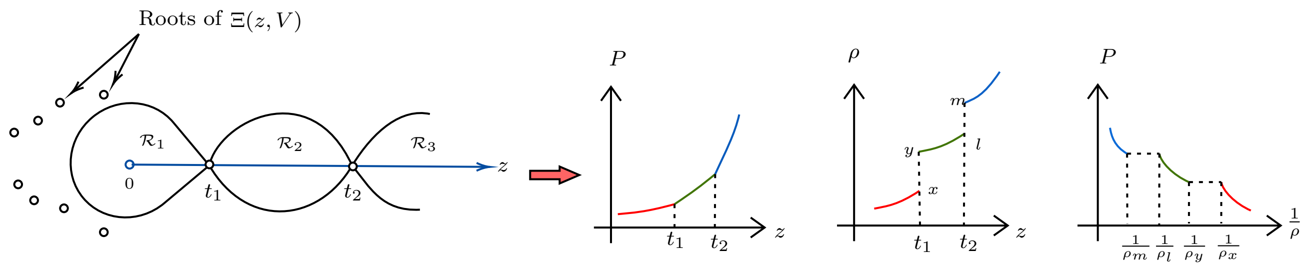
$$\rho = \lim_{V\to\infty} \frac{z}{V}\frac{\partial \ln \Xi}{\partial z} \iff z\frac{\partial}{\partial z}\left(\lim_{V\to\infty}\frac{1}{V}\ln \Xi\right) = z\frac{\partial(\beta P)}{\partial z} \tag{14}$$

If such a region $\mathcal{R}$ does not exist, then the limit does not exist which makes sense physically, since the density ρ as defined in Eq (13), does not assume a single value at the condensation point. Hence, everything depends on the existence of $\mathcal{R}$

- As $V\to\infty$, if a region $\mathcal{R}$ exists which fully encloses the positive real axis, then P exists from Thm (1) and is an analytic, increasing function of z. From Thm (2), ρ also exists and is an analytic function of z. It can be shown that ρ is also increasing with z. P and ρ are related by Eq (14). The system under consideration is in a *single phase*.



- On the other hand, if the roots of $\Xi(z,V)$ approach some specific values $z = t_1, t_2\ldots$ on the real axis as $V\to\infty$, from the figure below, there will exist regions $\mathcal{R}_1, \mathcal{R}_2\ldots$ within which Thm (2) will be satsified piecewise and hence, within these regions, the system exists in a single phase as argued before. At the points $t_1, t_2\ldots$, the pressure P will be continuous from Thm (1), but ρ might have some non-analyticity since this does not satisfy the condition for Thm (2). The system now possess *multiple phases*.



It is to be noted that the discontinuity in the density is only possible in the thermodynamic limit. For finite system, we will get a *continuous crossover* due to *finite-size effects*.

If $\frac{\partial P}{\partial z}$ is discontinuous, then we have a *first-order phase transition* as shown in the figure above. If $\frac{\partial P}{\partial z}$ is continuous but $\frac{\partial^2 P}{\partial z^2}$ is discontinuous, then we obtain a *second-order phase transition*.

From this discussion, we observe that the study of phase transitions can be reduced to just studying the distribution of the zeroes of the grand partition function.

Lecture 05: More on Van der Waal's Equation

In the previous section, we saw a mathematical theory of condensation based on the zeroes of the grand partition function. Let us now move back to the Van der Waal's equation and try to analyse the critical point.

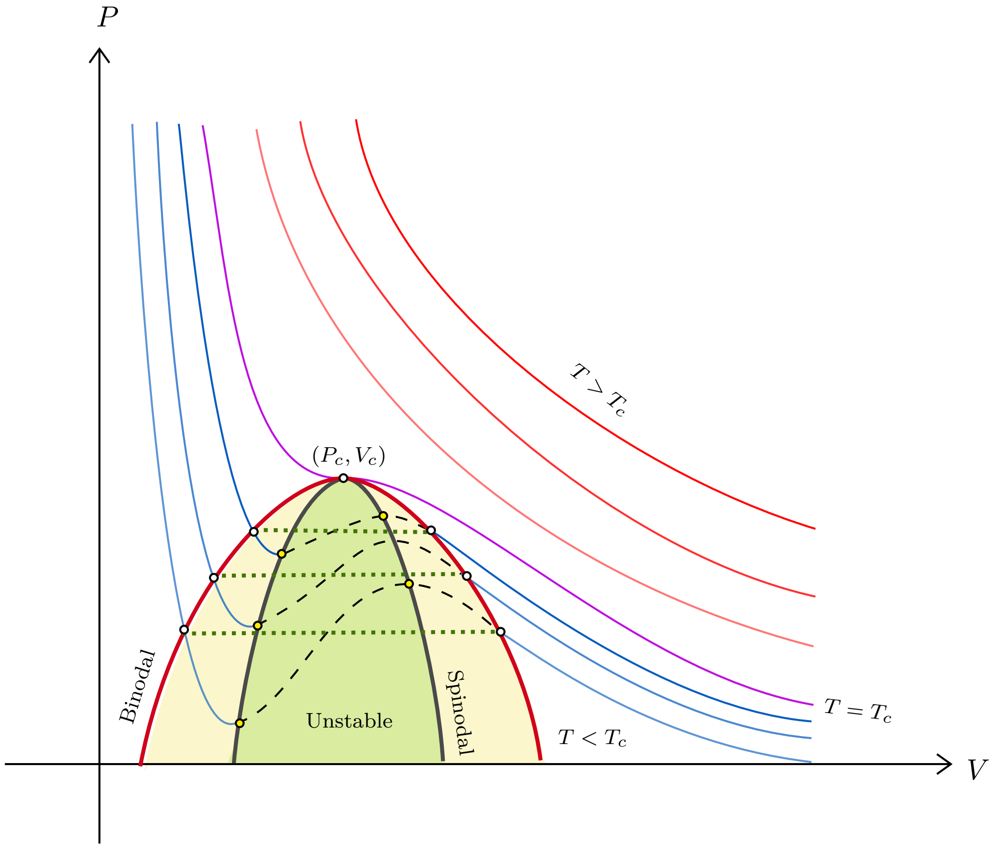


Figure 7: Van der Waal's isotherm. The figure shows Van der Waal's isotherms for $T < T_c$ ($T > T_c$) with blue (red) lines. The curve for $T = T_c$ is shown with purple line. Maxwell's construction is shown with green dotted line. The red line denotes the *binodal line* and the gray line denotes the *spinodal* line. The yellow region denote the metastable state while the green region denote the unstable states.

In Fig.??, we see the Van der Waal's isotherms at different temperature regime. There exists a temperature T_c such that at a particular point (V_c, P_c), the isotherm has zero slope and zero curvature. Thus the critical point is characterised by

$$\frac{\partial P}{\partial V} = 0 \qquad \frac{\partial^2 P}{\partial V^2} = 0$$

Defining $v = \frac{V}{N}$, from Van der Waal's equation we can write,

$$P = \frac{k_{\mathrm{B}}T}{v-b} - \frac{a}{v^2} \implies \frac{\partial P}{\partial V} = -\frac{k_{\mathrm{B}}T}{(v-b)^2} + \frac{2a}{v^3} \implies \frac{\partial^2 P}{\partial V^2} = \frac{2k_{\mathrm{B}}T}{(v-b)^3} - \frac{6a}{v^4} \tag{15}$$

Equating the derivative to zero, we get $2a(v_c - b)^2 = k_{\mathrm{B}}Tv_c^3$ and equating the second derivative to zero, we get $3a(v_c - b)^3 = k_{\mathrm{B}}Tv_c^4$. Dividing both these equations, we get

$$\frac{3}{2}(v_c - b) = v_c \implies 3(v_c - b) = 2v_c \implies v_c = 3b$$

Substituting this in any one of the above equation we get $k_{\mathrm{B}}T_c = \frac{8a}{27b}$ and substituting v_c and T_c in Van der Waal's equation, we get $P_c = \frac{a}{27b^2}$. Thus our critical parameters are,

$$\boxed{P_c = \frac{a}{27b^2} \qquad v_c = 3b \qquad k_{\mathrm{B}}T_c = \frac{8a}{27b}}$$

5.1. Approaching Universality

The Van der Waal's equation depends on the microscopic details of the system, a and b. These vary for different gases and hence, the equation cannot describe universal properties. However, let us define the *reduced parameters*,

$$\Pi = \frac{P}{P_c} \qquad \nu = \frac{v}{v_c} \qquad \tau = \frac{T}{T_c}$$

We now put these in the equation of state, which gives us

$$\begin{aligned}\left(\Pi P_c + \frac{a}{\nu^2 v_c^2}\right)(\nu v_c - b) = k_{\mathrm{B}}\tau T_c &\implies \left(\frac{\Pi a}{27b^2} + \frac{a}{9\nu^2 b^2}\right)(3b\nu - b) = \frac{8a\tau}{27b} \\ &\implies \frac{3ab}{27b^2}\left(\Pi + \frac{3}{\nu^2}\right)\left(\nu - \frac{1}{3}\right) = \frac{8a\tau}{27b} \\ &\implies \boxed{\left(\Pi + \frac{3}{\nu^2}\right)\left(\nu - \frac{1}{3}\right) = \frac{8\tau}{3}}\end{aligned} \tag{16}$$

The above dimensionless Van der Waal's equation in terms of the reduced parameters, is independent of a and b and is thus *universal*, in the sense that, different gases are expected to follow the same equation. The generalisation is known as the *principle of corresponding states*, which says that all fluids when compared with respect to reduced parameters, have the same *compressibility factor*. The compressibility factor is defined as $Z = \frac{Pv}{k_{\mathrm{B}}T}$, with the critical value predicted by Van der Waal to be,

$$Z_c = \frac{P_c v_c}{k_{\mathrm{B}} T_c} = \frac{3ab}{27b^2 \times \frac{8a}{27b}} = \frac{3}{8} = 0.375$$

Experimentally, the critical value of the compressibility factor is found to be in the range $0.28 - 0.33$, thus, Van der Waal's equation does not provide an accurate quantitative estimate.

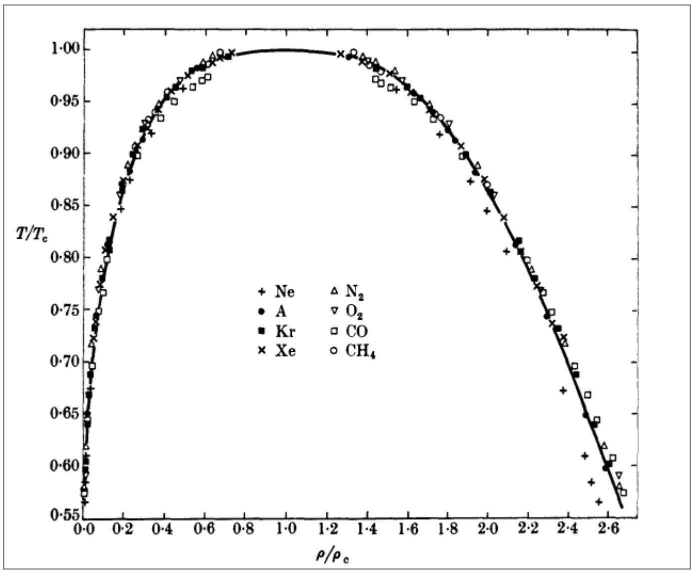


Figure 8: The Guggenheim plot for different gases with reduced parameter, showing that the trend follows a universal function (Source: Guggenheim (1945) [2]).

5.2. Critical Opalescence

Critical opalescence refers to an enhanced scattering of light by a substance in this critical region. Such enhanced scattering renders the substance milky white in reflected light.

As we know, at the critical point, $\mathcal{K}_{\mathrm{T}} = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T \to \infty$ since the slope of the isotherm is zero at the critical point. As $\mathcal{K}_{\mathrm{T}} \propto \sigma_{\mathrm{N}}$ where σ_{N} is the number fluctuations in the system, there is a huge number fluctuation, due to which the scattering of light is intensified.

5.3. Critical Point Behaviour

We will now discuss the behaviour of gases in the vicinity of the critical point to analyse universality. For $T > T_c$, there is always a unique volume corresponding to each pressure and hence, we can apply

Taylor expansion around v_c to write the pressure as,

$$P(T,v) = P(T,v_c) + \left.\frac{\partial P}{\partial v}\right|_T (v-v_c) + \frac{1}{2}\left.\frac{\partial^2 P}{\partial v^2}\right|_T (v-v_c)^2 + \frac{1}{6}\left.\frac{\partial^3 P}{\partial v^3}\right|_T (v-v_c)^3 + \mathcal{O}\left[\left(\frac{v-v_c}{v_c}\right)^4\right]$$

Each of the coefficients are functions of temperature and hence, can be Taylor expanded about T_c.

$$\begin{aligned}
P(T,v_c) &= \underbrace{P(T_c,v_c)}_{P_c} + \alpha(T-T_c) + \mathcal{O}\left[\left(\frac{T-T_c}{T_c}\right)^2\right] \\
\left.\frac{\partial P}{\partial v}\right|_{T,v_c} &= \cancelto{0}{\left.\frac{\partial P}{\partial v}\right|_{T_c,v_c}} - a(T-T_c) + \mathcal{O}\left[\left(\frac{T-T_c}{v_c}\right)^2\right] \\
\left.\frac{\partial^2 P}{\partial v^2}\right|_{T,v_c} &= \cancelto{0}{\left.\frac{\partial^2 P}{\partial v^2}\right|_{T_c,v_c}} + b(T-T_c) + \mathcal{O}\left[\left(\frac{T-T_c}{T_c}\right)^2\right] \\
\left.\frac{\partial^3 P}{\partial v^3}\right|_{T,v_c} &= -c + \mathcal{O}\left[\left(\frac{T-T_c}{T_c}\right)\right]
\end{aligned}$$

At the critical point the slope and the curvature vanishes and hence the first terms in the second and third equation is zero. Using the stability condition in Eq. (6), $a > 0$ for $T > T_c$ and $c > 0$. b can have any arbitrary sign[1]. Let us put these expression in the original Taylor expansion.

$$P(T,v) = P_c + \alpha(T-T_c) - a(T-T_c)(v-v_c) + \frac{b}{2}(T-T_c)(v-v_c)^2 - \frac{c}{6}(v-v_c)^3 + \text{higher orders}\ldots \tag{17}$$

From this, we can predict the following things about different quantities:

- Let us consider only upto linear order in $(T-T_c)$ and $(v-v_c)$, hence we neglect the fourth, fifth and higher order terms. This gives us an approximate analytic solution for the pressure,

$$\boxed{P(T,v) = P_c + \alpha(T-T_c) - a(T-T_c)(v-v_c)}$$

  We obtain $\frac{\partial P}{\partial v} = -a(T-T_c)$ and hence the isothermal compressibility at v_c scales as,

$$\lim_{T\to T_c^+} \mathcal{K}(T,v_c) = -\frac{1}{v_c}\frac{\partial v}{\partial P} \sim \frac{1}{T-T_c}$$

  Experimental results identify the exponent as $\gamma \approx 1.3$ where $\mathcal{K}(T,v_c) \sim (T-T_c)^{-\gamma}$.

- At $T = T_c$, the isotherm behaves as

$$P = P_c - \frac{c}{6}(v-v_c)^3 \implies (P-P_c) \sim (v-v_c)^3$$

  Experimental results identify the exponent as $\delta \approx 5.0$ where $(P-P_c) \sim (v-v_c)^\delta$.

The above analysis cannot be applied for $T < T_c$ since then for each pressure, we have two volumes and Taylor expansion cannot be effectively applied. Let us do the analysis in an alternate way using Eq. (16). Consider the two volumes v_{liq} and v_{gas} for which the pressure is same at say some temperature T. Then, from the equation we can write,

$$\Pi = \frac{8\tau/3}{\nu_g - 1/3} - \frac{3}{\nu_g^2} = \frac{8\tau/3}{\nu_l - 1/3} - \frac{3}{\nu_l^2}$$

[1] These a, b are not the ones appearing in the Van der Waal's equation.

From this we obtain,

$$8\tau\left[\frac{1}{\nu_g-1}-\frac{1}{\nu_l-1}\right]=3\left[\frac{1}{\nu_l^2}-\frac{1}{\nu_g^2}\right]=3\left[\frac{\nu_g^2-\nu_l^2}{\nu_l^2\nu_g^2}\right] \implies \boxed{\tau=\frac{(3\nu_l-1)(3\nu_g-1)(\nu_l+\nu_g)}{8\nu_g^2\nu_l^2}}$$

Note that the equation is symmetric in ν_g and ν_l and $\nu_g=\nu_l=1$ at the critical point, which implies that we can write

$$\nu_g=1+\frac{\varepsilon}{2} \qquad \nu_l=1-\frac{\varepsilon}{2}$$

where $\varepsilon\equiv\nu_g-\nu_l$. We substitute this in the expression for τ, leading us to

$$\tau=\frac{2\left(2-\frac{3\varepsilon}{2}\right)\left(2+\frac{3\varepsilon}{2}\right)}{8\left[\left(1+\frac{\varepsilon}{2}\right)\left(1-\frac{\varepsilon}{2}\right)\right]^2}=\frac{\left(4-\frac{9\varepsilon^2}{4}\right)}{4\left(1-\frac{\varepsilon^2}{4}\right)^2}=\left(1-\frac{9\varepsilon^2}{16}\right)\left(1-\frac{\varepsilon^2}{4}\right)^{-2}\approx\left(1-\frac{9\varepsilon^2}{16}\right)\left(1+\frac{\varepsilon^2}{2}\right)=1-\frac{\varepsilon^2}{16}+\mathcal{O}(\varepsilon^4)$$

Then substituting for the real variables, we have

$$(v_{\text{gas}}-v_{\text{liq}})\sim(T_c-T)^{1/2}$$

Experimental results predict an exponent $\beta\approx 0.3$ where $(v_{\text{gas}}-v_{\text{liq}})\sim(T_c-T)^{\beta}$.

Van der Waal's equation thus does not provide a perfect quantitative description of condensation, however, qualitative descriptions are more or less accurate. The exponents $\beta,\gamma,\delta\ldots$ are called *critical exponents* and many people dedicate their lives in finding the numerical value of these exponents.

Lecture 06: Correlations and Fluctuations

In here, we discuss a bit about an important quantity, *the correlation function*, which is an important quantity to look at while studying phase transition. Recall from Eq. (1) that the density was given by a sum over delta functions. We define the *density-density correlation function* (or simply the correlation function) as,

$$\mathcal{G}(\mathbf{r},\mathbf{r}')=\Big\langle[n(\mathbf{r})-\langle n(\mathbf{r})\rangle][n(\mathbf{r}')-\langle n(\mathbf{r}')\rangle]\Big\rangle=\langle n(\mathbf{r})n(\mathbf{r}')\rangle-\langle n(\mathbf{r})\rangle\langle n(\mathbf{r}')\rangle$$

where $n(\mathbf{r})-\langle n(\mathbf{r})\rangle$ denotes the fluctuation of the density from its average value and $\langle\cdot\rangle$ represent the *thermal (ensemble) average*, using the distribution. For translationally invariant system, we have $\mathcal{G}(\mathbf{r},\mathbf{r}')\equiv\mathcal{G}(\mathbf{r}-\mathbf{r}')$ and also, $n(\mathbf{r})=n(\mathbf{r}')$. Then the above equation can also be written as,

$$\mathcal{G}(\mathbf{r}-\mathbf{r}')=\langle n(\mathbf{r})n(\mathbf{r}')\rangle-n^2$$

where $\langle n(\mathbf{r})\rangle=n$. As $|\mathbf{r}-\mathbf{r}'|\to\infty$, that is, the distance between two particles become very large, we do expect them to be independent and no longer 'correlated'. Thus, $\mathcal{G}(\mathbf{r}-\mathbf{r}')\to 0$ as $|\mathbf{r}-\mathbf{r}'|\to\infty$.

The correlation function kind of quantifies the fluctuations in the density. Earlier, we saw that the isothermal compressibility also characterises the number fluctuations. Then, these two must be related on some level. To see this, we note that

$$\begin{aligned}\Big\langle(N-\langle N\rangle)^2\Big\rangle&=\Big\langle\int \mathrm{d}^3r\ \{n(\mathbf{r}-\langle n(\mathbf{r})\rangle)\}\int \mathrm{d}^3r'\ \{n(\mathbf{r}'-\langle n(\mathbf{r}')\rangle)\}\Big\rangle\\&=\int \mathrm{d}^3r\mathrm{d}^3r'\ \Big\langle\{n(\mathbf{r}-\langle n(\mathbf{r})\rangle)\}\{n(\mathbf{r}'-\langle n(\mathbf{r}')\rangle)\}\Big\rangle=\int\int \mathrm{d}^3r\mathrm{d}^3r'\ \mathcal{G}(\mathbf{r}-\mathbf{r}')\end{aligned}$$

Introducing the variables $\mathbf{r}''=\mathbf{r}-\mathbf{r}'$ and $\mathbf{R}=\mathbf{r}+\mathbf{r}'$, the integral changes accordingly.

$$\Big\langle(N-\langle N\rangle)^2\Big\rangle=\int \mathrm{d}^3R\int \mathrm{d}^3r''\mathcal{G}(r'')=V\int \mathrm{d}^3r''\mathcal{G}(r'')$$

The number fluctuation can also be written as,

$$\Big\langle(N-\langle N\rangle)^2\Big\rangle=\frac{\langle N\rangle^2 k_{\mathrm{B}}T}{V}\mathcal{K}_{\mathrm{T}}=\langle N\rangle nk_{\mathrm{B}}T\mathcal{K}_{\mathrm{T}}$$

Consider an ideal gas, with $PV = Nk_{\mathrm{B}}T$ which gives us,

$$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{Nk_{\mathrm{B}}T}{V^2} \implies \mathcal{K}_{\mathrm{T}}^{\circ} = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right) = \frac{V}{Nk_{\mathrm{B}}T} = \frac{1}{nk_{\mathrm{B}}T}$$

Using this, we can write the dimensionless compressiblity as,

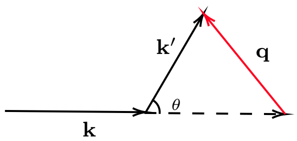
$$\left(\frac{\mathcal{K}_{\mathrm{T}}}{\mathcal{K}_{\mathrm{T}}^{\circ}}\right) = \frac{\left\langle (N - \langle N \rangle)^2 \right\rangle}{\langle N \rangle} = \frac{V \int \mathrm{d}^3 r\ \mathcal{G}(r)}{\langle N \rangle} = \frac{1}{n}\int \mathrm{d}^3 r\ \mathcal{G}(r)$$

The above is a very simple example of the *fluctuation-dissipation theorem* which relates the correlation to the linear response of the system (compressibility, in our case).

Since the isothermal compressibility diverges at T_c, the integral also diverges and this implies that the correlation function also becomes long ranged. The correlation length ξ (the characteristic length scale after which $\mathcal{G}(\mathbf{r})$ decays) becomes as large as the wavelength of light and the density fluctuations scatter light strongly, resulting in critical opalescence, as mentioned in sec. 5.2.

6.1. Structure Factor

Let us now try to quantify the scattering intensity when our fluid is subjected to some radiation. Assume a *quasielastic scatetring*, that is, the incoming radiation has much larger energy than the typical excitation energy of the system. Hence, the system can only absorb or emit very small amounts of energy and very small energy transfer occurs.



From elastic scattering, $|\mathbf{k}| \approx |\mathbf{k}'|$, the momentum transfer vector $\mathbf{q} = \mathbf{k}' - \mathbf{k}$ has a magnitude,

$$|\mathbf{q}| = \sqrt{|\mathbf{k}|^2 + |\mathbf{k}'|^2 - 2|\mathbf{k}||\mathbf{k}'|\cos\theta} \approx \sqrt{2|\mathbf{k}|^2(1-\cos\theta)} = 2|\mathbf{k}|\sin\left(\frac{\theta}{2}\right)$$

Now, suppose that there are N particles in the target and the *scattering amplitude* for scattering from particle i is $a_i(\mathbf{q})$. Then the total intensity of the scattered radiation is given by,

$$\mathcal{I}(q) = \left\langle \left| \sum_{j=1}^{N} a_j(\mathbf{q}) \right|^2 \right\rangle$$

The scattering amplitude for particles at $\mathbf{r}_i$ and $\mathbf{r}_j$ are related just by a phase factor,

$$a_j(\mathbf{q}) = a_i(\mathbf{q}) e^{-i\mathbf{q}\cdot(\mathbf{r}_j - \mathbf{r}_i)}$$

Using this, we can write the internsity as,

$$\mathcal{I}(\mathbf{q}) = \left\langle \left| \sum_{j=1}^{N} a_1(\mathbf{q}) e^{-i\mathbf{q}\cdot(\mathbf{r}_j - \mathbf{r}_1)} \right|^2 \right\rangle = |a_1(\mathbf{q})|^2 \left\langle \left| \sum_{j=1}^{N} e^{-i\mathbf{q}\cdot(\mathbf{r}_j - \mathbf{r}_1)} \right|^2 \right\rangle = |a_1(\mathbf{q})|^2 \left\langle \left| \sum_{j=1}^{N} e^{-i\mathbf{q}\cdot\mathbf{r}_j} \right|^2 \right\rangle \tag{18}$$

as $\left|e^{i\mathbf{q}\cdot\mathbf{r}_1}\right| = 1$. If no correlations exist between the particles, then $\sum\limits_j e^{-i\mathbf{q}\cdot(\mathbf{r}_j - \mathbf{r}_1)} = N + \sum\limits_{j\neq 1} e^{-i\mathbf{q}\cdot(\mathbf{r}_j - \mathbf{r}_1)}$ and $\sum\limits_{j\neq 1} \left\langle e^{-i\mathbf{q}\cdot(\mathbf{r}_j - \mathbf{r}_1)} \right\rangle \approx 0$ due to randomised phases which leads to,

$$\mathcal{I}^{\circ}(\mathbf{q}) = N|a_1(\mathbf{q})|^2$$

Then we can write the dimensionless intensity as,

$$\begin{aligned}
\frac{\mathcal{I}(\mathbf{q})}{\mathcal{I}^{\circ}(\mathbf{q})} &= \frac{1}{N}\left\langle\left(\sum_{j=1}^{N} e^{i\mathbf{q}\cdot\mathbf{r}_j}\right)\left(\sum_{i=1}^{N} e^{-i\mathbf{q}\cdot\mathbf{r}_i}\right)\right\rangle = \frac{1}{N}\left\langle\left(\sum_{i,j} e^{i\mathbf{q}\cdot(\mathbf{r}_j-\mathbf{r}_i)}\right)\right\rangle \\
&= \frac{1}{N}\int \mathrm{d}^3r\mathrm{d}^3r' \ \left\langle\left(\sum_{i,j}\delta^{(3)}(\mathbf{r}-\mathbf{r}_i)\delta^{(3)}(\mathbf{r}'-\mathbf{r}_j)e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')}\right)\right\rangle \\
&= \frac{1}{N}\int \mathrm{d}^3r\mathrm{d}^3r' \ \left\langle e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')}\left(\sum_{i,j}\delta^{(3)}(\mathbf{r}-\mathbf{r}_i)\delta^{(3)}(\mathbf{r}'-\mathbf{r}_j)\right)\right\rangle \\
&= \frac{1}{N}\int \mathrm{d}^3r\mathrm{d}^3r' \ e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')}\left\langle\left(\sum_{i}\delta^{(3)}(\mathbf{r}-\mathbf{r}_i)\right)\left(\sum_{j}\delta^{(3)}(\mathbf{r}'-\mathbf{r}_j)\right)\right\rangle \\
&= \frac{1}{N}\int \mathrm{d}^3r\mathrm{d}^3r' \ e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')}\left\langle n(\mathbf{r})n(\mathbf{r}')\right\rangle \\
&= \frac{1}{N}\int \mathrm{d}^3r\mathrm{d}^3r' \ e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')}(\mathcal{G}(\mathbf{r}-\mathbf{r}')+n^2)
\end{aligned}$$

Again using the same variable substitution, we obtain

$$\frac{\mathcal{I}(\mathbf{q})}{\mathcal{I}^{\circ}(\mathbf{q})} = \frac{V}{N}\int \mathrm{d}^3r\ e^{-i\mathbf{q}\cdot\mathbf{r}}\mathcal{G}(\mathbf{r}) + \frac{V^2}{N}\int \mathrm{d}^3r\ e^{-i\mathbf{q}\cdot\mathbf{r}} = \frac{V}{N}\int \mathrm{d}^3r\ e^{-i\mathbf{q}\cdot\mathbf{r}}\mathcal{G}(\mathbf{r}) + \frac{V^2n^2}{N}\delta^{(3)}(\mathbf{q})$$

The last term contributes only when $\mathbf{q} = 0$ which is for $\theta = 0$ (forward-scattering) and is omitted in most cases. Hence, we define the structure factor $\mathcal{S}(\mathbf{q})$ as

$$\mathcal{S}(\mathbf{q}) = \frac{n\mathcal{I}(\mathbf{q})}{\mathcal{I}^{\circ}(\mathbf{q})} = \int \mathrm{d}^3r\ e^{-i\mathbf{q}\cdot\mathbf{r}}\mathcal{G}(\mathbf{r})$$

$\mathcal{S}(\mathbf{q})$ is just the spatial Fourier transform of the correlation function. $\mathcal{S}(\mathbf{q})$ becomes extremely large as we approach the critical point, for small values of $|\mathbf{q}|$.

Lecture 07: Introduction to Ising Model

The Ising model is a prototypical model to study phase transitions in magnetic systems. Since its fomulation by *Lenz* in 1920 and subsequent 1D solution by *Ising* and 2D solution by *Onsager*, Ising model has penetrated deep into all fields. The Ising model is defined faily simply, by the Hamiltonian,

$$\mathcal{H} = -J\sum_{\langle ij\rangle}\sigma_i\sigma_j - h\sum_j \sigma_j \tag{19}$$

where $\sigma_j \in \{+1, -1\}$ is a random variables which is given an interpretation of a *classical spin*. J and h respectively denote the spin-coupling constant and the external field. The notation $\langle ij\rangle$ means that the spins i and j are nearest-neighbours. The Ising model provides interesting numerical and analytical artefacts till date which makes it an important system to study. The 1D solution is fairly simple, using a *transfer-matrix* method which is provided in appendix B.

7.1. Mean-field solution

In the mean-field solution for the Ising model, we 'throw' away the fluctuations in each spin, which makes the spins interact with an 'effective' field resulting from the average orientation of the neighboring spins. Consider the identity,

$$(\sigma_i - \langle\sigma_i\rangle)(\sigma_j - \langle\sigma_j\rangle) = \sigma_i\sigma_j - \sigma_i\langle\sigma_j\rangle - \sigma_j\langle\sigma_i\rangle + \langle\sigma_i\rangle\langle\sigma_j\rangle \tag{20}$$

The LHS is zero since we are neglecting the spin fluctuations in the mean-field and hence the Ising Hamiltonian in Eq. (19) becomes,

$$\mathcal{H} = -J\sum_{\langle ij\rangle}\sigma_i\langle\sigma_j\rangle - J\sum_{\langle ij\rangle}\sigma_j\langle\sigma_j\rangle + J\sum_{\langle ij\rangle}\langle\sigma_i\rangle\langle\sigma_j\rangle - h\sum_i\sigma_i$$

Due to translation invariance, $\langle\sigma_i\rangle = \langle\sigma_j\rangle \equiv \langle\sigma\rangle$ and hence can be taken out of the sum.

$$\begin{aligned}\mathcal{H} &= -J\langle\sigma\rangle\sum_{\langle ij\rangle}\sigma_i - J\langle\sigma\rangle\sum_{\langle ij\rangle}\sigma_j + J\langle\sigma\rangle^2\sum_{\langle ij\rangle} - h\sum_i\sigma_i\\ &= -2J\langle\sigma\rangle\sum_{\langle ij\rangle}\sigma_i + J\langle\sigma\rangle^2\sum_{\langle ij\rangle} - h\sum_i\sigma_i\end{aligned}$$

For each spin σ_i, let us suppose that we have γ neighbours. Since the sums are now independent of j, these can be reduced to a sum over spins,

$$\sum_{\langle ij\rangle} f(\sigma_i) = \frac{1}{2}\sum_i f(\sigma_i)\sum_{j\in\mathrm{NN}(i)} \equiv \frac{\gamma}{2}\sum_{i=1}^{N} f(\sigma_i) \implies \begin{aligned}\mathcal{H} &= -J\gamma\langle\sigma\rangle\sum_i\sigma_i + \frac{JN\gamma}{2}\langle\sigma\rangle^2 - h\sum_i\sigma_i\\ &= -\mathrm{h}_{\mathrm{eff}}\sum_i\sigma_i + \frac{JN\gamma}{2}\langle\sigma\rangle^2\end{aligned} \tag{21}$$

where $\mathrm{h}_{\mathrm{eff}} = h + J\gamma\langle\sigma\rangle$ is kind of the *mean field* which is subjected to each spin. The Hamiltonian depends on $\langle\sigma\rangle$, however, given the Hamiltonian, we can also calculate $\langle\sigma\rangle$ from the partition function. This leads to a *self-consistent* equation which we need to tackle!

Lecture 08: Mean-field solution for Ising model

We will continue our discussion of the mean-field solution of the Ising model. Previously, we had reduced the Hamiltonian to a simplified expression, with an effective field.

$$\mathcal{H}_{\mathrm{MF}} = -\mathrm{h}_{\mathrm{eff}}\sum_i\sigma_i + \frac{JN\gamma}{2}\langle\sigma\rangle^2 \tag{22}$$

This is simple to solve, since this is N independent spins in a field. The partition function is,

$$\mathcal{Z} = \sum_{\{\sigma_i\}} e^{\beta\mathrm{h}_{\mathrm{eff}}\sum\sigma_i}\cdot e^{-\beta\mathcal{E}_0} \tag{23}$$

where $\mathcal{E}_0 = \frac{J\gamma N}{2}\langle\sigma\rangle^2$. Since the particles are independent, the sum over spins can be simplified as,

$$\mathcal{Z} = \left[\sum_{\sigma\in\{-1,1\}} e^{\beta\mathrm{h}_{\mathrm{eff}}\sigma}\right]^N\cdot e^{-\beta\mathcal{E}_0} = [2\cosh(\beta\mathrm{h}_{\mathrm{eff}})]^N e^{-\beta\mathcal{E}_0}$$

From this, the free energy is obtained as,

$$F(T,h) = -k_{\mathrm{B}}T\ln\mathcal{Z} = k_{\mathrm{B}}T[\beta\mathcal{E}_0 - N\ln(2\cosh\beta\mathrm{h}_{\mathrm{eff}})] = \mathcal{E}_0 - Nk_{\mathrm{B}}T\ln(2\cosh\beta\mathrm{h}_{\mathrm{eff}})$$

From the free-energy, we can obtain the magnetisation,

$$\mathcal{M} = -\frac{\partial F}{\partial B} = Nk_{\mathrm{B}}T\tanh\beta\mathrm{h}_{\mathrm{eff}}\times\beta = N\tanh\beta\mathrm{h}_{\mathrm{eff}}$$

The magnetisation per spin is thus obtained as a *transcendental equation*,

$$\boxed{\langle\sigma\rangle = \frac{\mathcal{M}}{N} = \tanh(\beta J\gamma\langle\sigma\rangle + \beta h)} \tag{24}$$

We are interested in finding whether we have any spontaneous magnetisation in the system, that is, magnetisation without an external field. Thus, let us take the case when $h = 0$. Now, define $x := \frac{\langle\sigma\rangle}{\beta J\gamma}$, so that the equation changes to,

$$\frac{x}{\beta J\gamma} = \tanh x$$

Now, note that $\beta J\gamma$ should be dimensionless as x is dimensionless. This implies that $J\gamma$ should have dimension of $k_\mathrm{B}T$. Let us define $k_\mathrm{B}T_c := J\gamma$ which leads us to,

$$\left(\frac{T}{T_c}\right)x = \tanh(x)$$

Given a temperature T, the left-hand side is just an equation of a line, with slope $\frac{T}{T_c}$. Let us analyse the situation graphically.

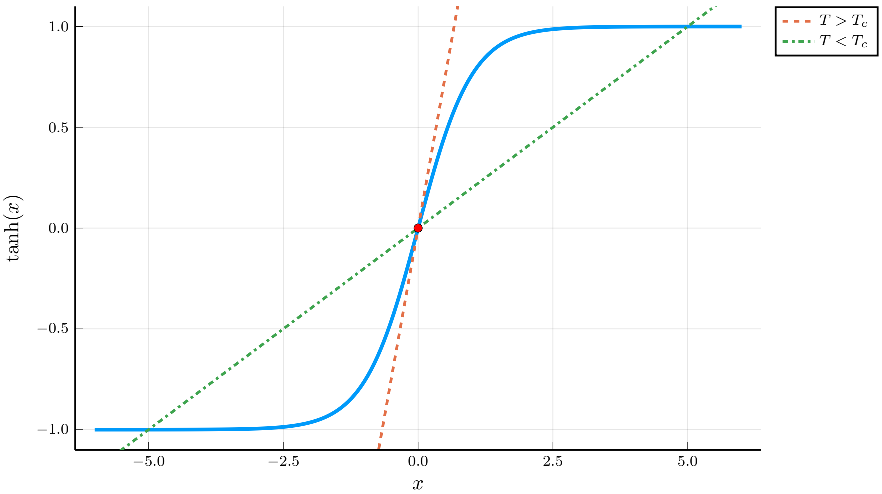


Figure 9: Graphical analysis of the transcendental equation

In Fig.9, we have plotted $\tanh(x)$ and two straight lines, with $T > Tc$ and $T < T_c$. For $T > Tc$, we see that the line does not cut $\tanh(x)$ at any point other than $x = 0$. However, for $T < T_c$, we see that two solutions exist, other than the trivial solution $x = 0$. As $\tanh(-x) = -\tanh(x)$, if one of the solution is $x = +x_0$, then the other solution is $x = -x_0$. A solution of x corresponds to a solution of $\langle\sigma\rangle$. Thus,

- $T > T_c$: $\langle\sigma\rangle = 0$ is the only solution and there is no spontaneous magnetisation.
- $T < T_c$: Apart from $\langle\sigma\rangle = 0$, we have two other solutions $\langle\sigma\rangle = \pm m_0$.

The mean-field solution, which predicts a finite critical temperature, is clearly wrong for the 1D case, where we know that no phase transition occur at any finite temperature. In 1D, the fluctuations are too strong to throw away, since there are only two nearest neighbours to interact with.

For the Ising model, two important things to note are *dimensionality* and lattice structure. However, in the mean-field Hamiltonian, γ (which is the only parameter depending on the topology of the lattice) cannot distinguish between these two factors. For example, for a cubic lattice in 3D, we have six nearest neighbours. Also, a triangular lattice in 2D has six nearest neighbours. Hence, these two lattices have the same γ, however, Ising dynamics in both the lattices are expected to be quite different.

Mean-field analysis becomes exact if we increase the dimensionality of the lattice and also if we increase the range of interaction, that is, we include higher order neighbours also. This ensures that the interaction between the spins is indeed an *effective interaction*.

Lecture 09: More on Ising Model

We have predicted a phase transition for the Ising model using mean-field approximation. However, the approximation seems to be inconsistent on many grounds, working well for higher dimensions and long-range interactions. It predicted that in 1D too, Ising model has a spontaneous magnetisation at some finite temperature which cannot be. Even though we have solved the Ising system for dimension $d = 1$ in appendix. B (and it has been painfully solved by many stalwarts for $d = 2$ case too), let us see some arguments on why phase transitions cannot occur for 1D but it is possible in 2D Ising system.

9.1. Phase Transition in 1D Ising System

We will show that for $T > 0$ in $d = 1$ zero-field Ising Model, there exists no *long-range order*. Consider the ground state of the Ising model at $T = 0$, which is *ordered*, with all spins aligned in the same direction. Under open boundary conditions, this configuration has minimum energy $E_0 = -J(N - 1)$.

$$\dots \uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow \dots$$

The above configuration has entropy $S = 0$, since all the spins are aligned. Hence, the free energy of the system is $F_{\text{grnd}} = E - TS = -J(N - 1)$.

On increasing the temperature, the spins flips randomly due to interactions and this increases the energy of the system (since the interaction term is $-J(-1) = +J$ only if neighbouring spins are oppositely aligned). We need to check whether this destroys the long-range order. Now, consider a configuration with one *domain wall*[1] introduced at some point in the Ising chain, as shown in red below.

$$\dots \uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow \quad \downarrow\downarrow\downarrow\downarrow\downarrow\downarrow\downarrow\downarrow \dots$$

Introducing the domain wall increases the energy, due to the spins on either side of the domain wall.

$$E_{\text{dw}} = -J(N_+ - 1) - J(N_- - 1) - J(-1) = -J(N - 1) + 2J$$

Thus introducing one domain wall incurs a cost of $\Delta E = 2J$. Now, since there are N spins, this domain wall can be placed at any of the $(N - 1)$ positions. Then the entropy of the system with one domain wall is $S = k_{\text{B}} \ln(N - 1)$ and the free energy is $F_{\text{dw}} = -J(N - 1) + 2J - k_{\text{B}} T \ln(N - 1)$. The change in free energy is,

$$\Delta F = F_{\text{dw}} - F_{\text{grnd}} = 2J - k_{\text{B}} T \ln(N - 1)$$

In the thermodynamic limit $N \to \infty$ which implies that $\Delta F \to -\infty$. The system can always lower the free energy by creating a domain wall for any temperature $T > 0$. In fact, the free energy is further lowered by introducing arbitrary number of domain walls, untill no domains remain at all. This implies that the long-ranged order is unstable and there is no spontaneous magnetisation at any finite temperature in the zero-field $d = 1$ Ising model.

9.2. Phase Transition in 2D Ising System (Peierls' Argument)

The situation becomes a bit different for $d = 2$ Ising model. To prove the existence of phase transition at a finite temperature in $d = 2$, consider a case where all the outer layer spins in the lattice are of the same type, say $\uparrow$. Now, consider the spin in the center of the lattice, highlighted yellow in Fig. 10.

[1] In simple terms, a domain wall is an imaginary boundary which separates *domains*. Domains are regions with the same kind of spins, that is, all up or all down.

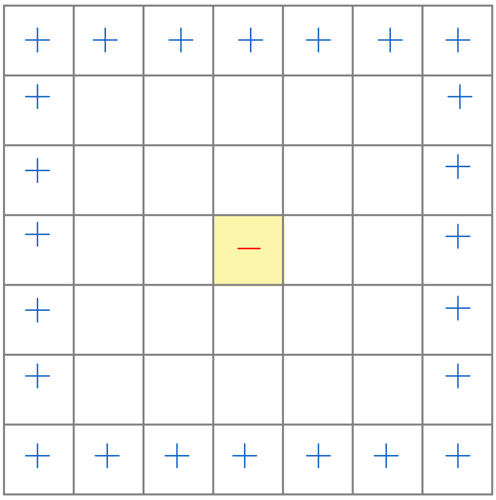
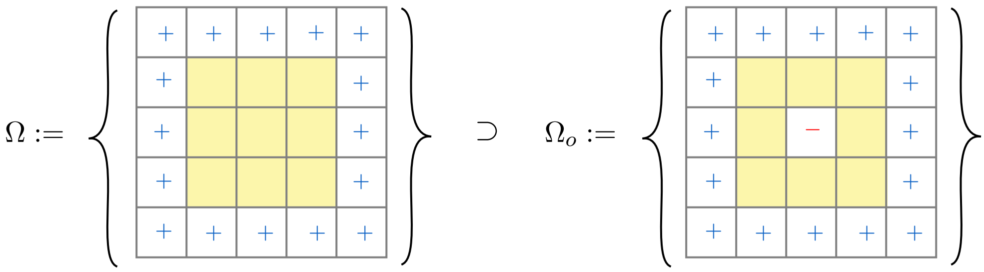


Figure 10: 2D Ising lattice; the outer layer have all same spins while the center spin is oppositely aligned with the outer spins.

If there is spontaneous magnetisation at some finite temperature, then the probability of the center spin to be oppositely aligned to the outer spins become zero (since the lattice becomes *ordered*). Also, note that the central spin is not particularly important and is just a representative of the bulk spin.



Now, consider Ω to be the set of all configurations where all the outer layers have spin $+$. Naturally, if we define Ω_0 to be the set of all configurations where the central spin is $-$ and all the outer layers have spin $+$, we have $\Omega_0 \subset \Omega$, as shown in the above figure. To have a phase transition is to show that the probability of any configuration $\{\sigma\} \in \Omega_0 \to 0$ as $N \to \infty$.

To show that, we define a *shoreline* which is an imaginary closed loop separating $+$ spins from $-$ spins. So, along the boundary of the shoreline, we have all opposite spins on either side. A given shoreline thus separates spins σ_i and σ_j such that $\sigma_i \cdot \sigma_j = -1$. Note that, in the interior of the shoreline, there can be $+$ or $-$ spins.

We define another set of configurations, Ω_s in which every configuration has the central spin to be $-$ and all the outer layers have spin $+$. In addition to it, there is a *fixed* shoreline S surrounding the central spin. Clearly, $\Omega_s \subset \Omega_0$

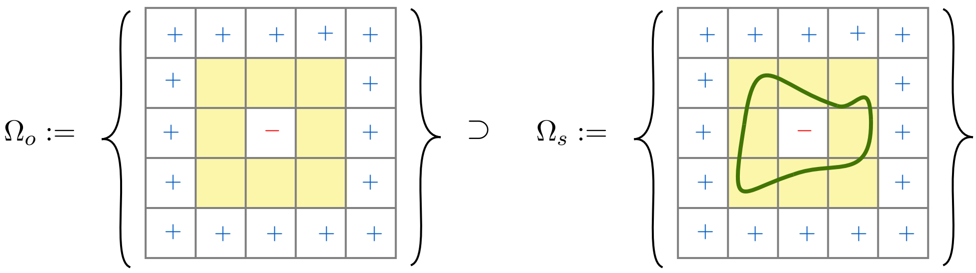


Figure 11: The set of configurations showing the shoreline with a green line.

The probability of a configuration inside Ω_s is given by,

$$\text{Prob}(\{\sigma\} \in \Omega_s) = \frac{1}{\mathcal{Z}} \sum_{\{\sigma\}\in\Omega_s} e^{-\beta\mathcal{H}(\{\sigma\})} = \frac{1}{\mathcal{Z}} \sum_{\{\sigma\}\in\Omega_s} \exp\left(\beta J \sum_{\langle ij\rangle\in S} \sigma_i\sigma_j\right) \exp\left(\beta J \sum_{\langle ij\rangle\notin S} \sigma_i\sigma_j\right) \tag{25}$$

where we have broken the nearest neighbour sum into two parts: spins which are separated by the shoreline and spins which are not. Note that if the spins are separated by the shoreline, then $\sigma_i\sigma_j = -1$ and the sum is just $-n(S)$, where $n(S)$ denotes the *length* of the shoreline. Thus, the probability becomes,

$$\text{Prob}(\{\sigma\} \in \Omega_s) = \frac{1}{\mathcal{Z}} \sum_{\{\sigma\}\in\Omega_s} e^{-\beta J n(S)} \exp\left(\beta J \sum_{\langle ij\rangle\notin S} \sigma_i\sigma_j\right) \tag{26}$$

The remaining sum, where spins are not separated by the shoreline, is hard to calculate exactly and we need to find an estimate for that.

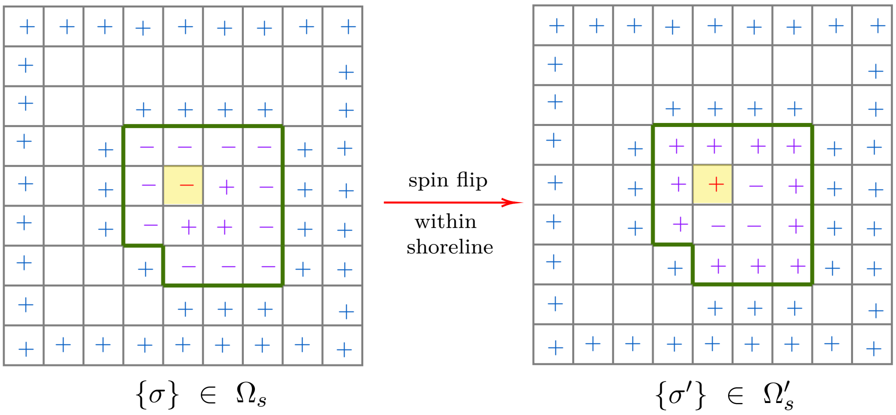


Figure 12: Flipping of all the spins inside the shoreline, taking $\sigma \to \sigma'$

We consider a specific configuration $\{\sigma\} \in \Omega_s$ and then flip all the spins in the inner side of the shoreline as shown in Fig. 12. We call this configuration $\{\sigma'\}$. If a spin in this configuration is outside the shoreline, then $\sigma_i = \sigma'_i$ since the spins outside the shoreline remains unchanged. If the spins belong inside the shoreline, then both spins have flipped and we have $\sigma'_i\sigma'_j = (-\sigma_i)(-\sigma_j) = \sigma_i\sigma_j$. If the spins are separated by shoreline, only one of the spins (which is in the inner side of the shoreline) changes and $\sigma'_i\sigma'_j = -\sigma_i\sigma_j = -(-1) = +1$. Thus we have,

$$\sum_{\langle ij\rangle} \sigma'_i\sigma'_j = \sum_{\langle ij\rangle\in S} \sigma'_i\sigma'_j + \sum_{\langle ij\rangle\notin S} \sigma'_i\sigma'_j = n(S) + \sum_{\langle ij\rangle\notin S} \sigma_i\sigma_j$$

Note that $n(S) > 0$ always which give us the inequality,

$$\sum_{\langle ij\rangle\notin S} \sigma_i\sigma_j = \sum_{\langle ij\rangle} \sigma'_i\sigma'_j - n(S) < \sum_{\langle ij\rangle} \sigma'_i\sigma'_j$$

Also note that $\sigma_i \to \sigma'_i$ is a one-to-one map and hence the sum of spins over Ω_s is equal to the sum of spins over Ω'_s. Using this and the estimate of the sum, Eq. (26) becomes,

$$\begin{aligned}\text{Prob}(\{\sigma\} \in \Omega_s) &< \frac{1}{\mathcal{Z}} \sum_{\{\sigma'\}\in\Omega'_s} e^{-\beta J n(S)} \exp\left[\beta J \sum_{\langle ij\rangle} \sigma'_i\sigma'_j\right] \\ &= \frac{e^{-\beta J n(S)}}{\mathcal{Z}} \sum_{\{\sigma'\}\in\Omega'_s} \exp\left[\beta J \sum_{\langle ij\rangle} \sigma'_i\sigma'_j\right]\end{aligned} \tag{27}$$

Now, the RHS is the sum over all configurations in Ω'_s. Since $e^x > 0$ for all x, this term is less than the sum over all the configurations $\{\sigma\} \in \Omega$. Thus,

$$\text{Prob}(\{\sigma\} \in \Omega_s) < e^{-\beta J n(S)} \frac{1}{\mathcal{Z}} \sum_{\{\sigma\}\in\Omega} \exp\left[\beta J \sum_{\langle ij \rangle} \sigma_i \sigma_j\right] = e^{-\beta J n(S)}$$

since the red term is just the definition of the partition function. Thus the probability of any configuration in Ω_s is less than $e^{-\beta J n(S)}$. However, we were interested in the situation where the configuration belonged to Ω_0 which can be obtained by summing over all such possible shorelines S.

$$\text{Prob}(\{\sigma\} \in \Omega_0) = \sum_{\{S\}} \text{Prob}(\Omega_S) < \sum_{\{S\}} e^{-\beta J n(S)}$$

Note that the length n of a shoreline can be $n = 4, 5, 6 \ldots$. The minimum value is 4, for the shoreline which contains only the central spin and this length increases. Then the sum over shorelines can be written as,

$$\text{Prob}(\{\sigma\} \in \Omega_0) < \sum_{n=4}^{\infty} s(n) e^{-\beta J n}$$

where $s(n)$ denote the number of shorelines with length n. The final thing to do is to have an estimate for $s(n)$. For this, note that shorelines are closed curves. To form a shoreline, start from any point P. We can move in only three directions (not in four) since shorelines are self-avoiding, so we cannot move in the direction in which we had already formed a line.

The number of ways to form a shoreline of length n is thus $\sim 3^n$. Note that in this case, the choice of P was arbitrary and we can essentially start from any of the n points. Thus, the number of shorelines of length n is $s(n) \sim 3^n n$. Substituting this in the above expression,

$$\text{Prob}(\{\sigma\} \in \Omega_0) < \sum_{n=4}^{\infty} 3^n n e^{-\beta J n} \equiv \sum_{n=4}^{\infty} n r^n < \sum_{n=1}^{\infty} n r^n \qquad \text{where, } r = 3e^{-\beta J}$$

From the ratio test, we have

$$\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty} \left|\frac{(k+1) r^{k+1}}{k r^k}\right| = \lim_{k\to\infty} \left(1 + \frac{1}{k}\right) |r| = |r|$$

and hence the sum converges if $|r| < 1 \implies e^{\beta J} > 3$. The value of the sum in the RHS is $\frac{r}{(r-1)^2}$ and hence,

$$\text{Prob}(\{\sigma\} \in \Omega_0) < \frac{r}{(r-1)^2} = \frac{3e^{-\beta J}}{(3e^{-\beta J} - 1)^2}$$

As $\beta \to \infty$ (corresponding to $T \to 0$), $\text{Prob}(\{\sigma\} \in \Omega_0) \to 0$ and hence for sufficiently low temperatures all spins in the systems tend to align even without an external magnetic field, proving the existence of a phase transition.

9.3. Symmetries of the Ising Model

The zero-field Ising Hamiltonian has an obvious symmetry of $\sigma_i \mapsto -\sigma_i$, which is broken if $h \neq 0$. In non-zero field, from Eq. (19), we can see that if $h \to -h$ and $\{\sigma\} \to \{-\sigma\}$, then the Ising Hamiltonian remains invariant.

$$\mathcal{H}(J, h, \{\sigma\}) = \mathcal{H}(J, -h, \{-\sigma\})$$

This is called the *time-reversal* symmetry or $\mathbb{Z}_2$ symmetry. Let us see what happens to the partition function under $h \to -h$.

$$\begin{aligned}
\mathcal{Z}(-h, J, T) &= \sum_{\{\sigma\}} \exp\left[-\beta \mathcal{H}(-h, J, \{\sigma\})\right] \\
&= \sum_{\{\sigma\}} \exp\left[-\beta \mathcal{H}(-h, J, \{-\sigma\})\right] \\
&= \sum_{\{\sigma\}} \exp\left[-\beta \mathcal{H}(h, J, \{\sigma\})\right] = \mathcal{Z}(h, J, T)
\end{aligned}$$

where in the second sum we used the fact that changing $\sigma \to -\sigma$ will not matter since we are summing over both -1 and $+1$ spins. The free energy is just the logarithm of the partition function, hence it also enjoys the symmetry,

$$F(-h, J, T) = F(h, J, T)$$

From the free energy, we obtain the magnetisation

$$\mathcal{M}(h) = -\frac{\partial F(h,J,T)}{\partial h} = -\frac{\partial F(-h,J,T)}{\partial h} = \frac{\partial F(-h,J,T)}{\partial(-h)} = -\mathcal{M}(-h) \tag{28}$$

From the above expression, if $h = 0$ then we have $\mathcal{M}(0) = 0$ which apparently says that *phase transition is impossible in zero-field!*

This situation arises since we assumed the derivative of the free energy is well-behaved which is indeed true for finite systems, however, in the thermodynamic limit, F develops a kink for $T < T_c$ and hence $\left(\frac{\partial F}{\partial h}\right)$ develops a discontinuity as shown in Fig. 13 (a)

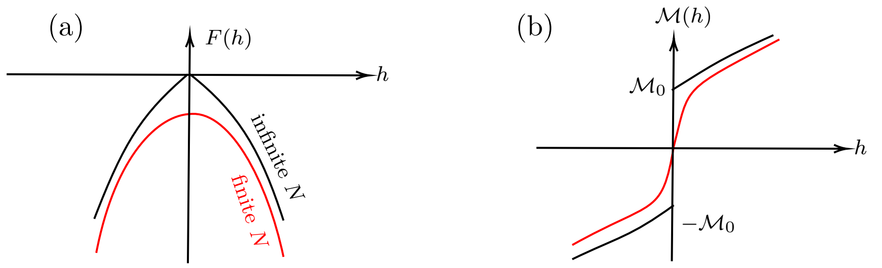


Figure 13: (a) Free energy and (b) Magnetisation behaviour in finite N and thermodynamic limit for $T < T_c$.

In the thermodynamic limit, near $h = 0$ the free energy behaves as,

$$F(h) = F(0) - \mathcal{M}_0|h| + \mathcal{O}(h^{q>1})$$

From this, if $h < 0$ then $\mathcal{M}(0^-) = -\mathcal{M}_0$ and if $h > 0$ then $\mathcal{M}(0^+) = +\mathcal{M}_0$ which is clearly discontinuous as shown in Fig. 13 (b). It is clear that if we take the thermodynamic limit first and then take $h \to 0$ and if we first take $h \to 0$ for a finite N and then take $N \to infty$, we get two different behaviours. Thus, the limits do not commute.

$$\lim_{N\to\infty} \lim_{h\to 0} \left(\frac{1}{N}\frac{\partial F(h)}{\partial h}\right) \neq \lim_{h\to 0} \lim_{N\to\infty} \left(\frac{1}{N}\frac{\partial F(h)}{\partial h}\right)$$

9.3.1. Spontaneous Symmetry Breaking

When the Hamiltonian of the system has a particular symmetry but the statistical behavior of the system breaks that symmetry, we say that the symmetry is broken spontaneously. As seen before, the zero-field Ising Hamiltonian enjoys a $\mathbb{Z}_2$ symmetry, under the exchange $\{\sigma_i\} \to \{-\sigma_i\}$. For $T < T_c$ in the thermodynamic limit, we saw that

$$m = \lim_{N\to\infty} \frac{1}{N} \sum_i \langle\sigma_i\rangle \neq 0$$

Hence, at all temperature $\mathcal{H}$ has a symmetry under the exchange of spins but $\langle\sigma_i\rangle$ does not. Thus we say that the symmetry is spontaneously broken.

This is different from the uninteresting explicit symmetry breaking in the $h \neq 0$ case, where the Hamiltonian itself does not have the $\mathbb{Z}_2$ symmetry from beginning.

Spontaneous symmetry breaking seems a bit counterintuitive if we naively see a probabilistic viewpoint of the situation. Since the Hamiltonian has $\mathbb{Z}_2$ symmetry, the Boltzmann factor $P(\{\sigma\}) =$

$\frac{1}{Z}\exp\left(-\beta\mathcal{H}(\{\sigma\})\right)$ is also invariant under this symmetry. Then the expectation,

$$\langle\sigma_i\rangle = \sum_{\{\sigma\}}\sigma_i P(\{\sigma\}) = 0$$

The sum is zero because for a configuration $\{\sigma_i\}$, the opposite configuration $\{-\sigma_i\}$ will have the same Boltzmann factor, however, the σ_i factor in the sum will become $-\sigma_i$, which will cancel with the configuration. Thus, in the thermodynamic limit, we have to be careful with the calculation!

Consider the Ising model when $h \neq 0$. In this case, consider two configurations A and B which differ by the fact that all spins in configuration B are the flipped spins of configuration A. Due to the field term $-h\sum\sigma_i = -hNm$, the probability of these two contributions will be

$$\frac{P_{\mathrm{A}}}{P_{\mathrm{B}}} = \frac{\exp[\beta hNm]}{\exp[-\beta hNm]} \implies \frac{P_{\mathrm{B}}}{P_{\mathrm{A}}} = \exp[-2\beta hNm]$$

In the thermodynamic limit, for $h > 0$, $\frac{P_{\mathrm{B}}}{P_{\mathrm{A}}} \to 0$ and hence the system must be in configuration A, with magnetisation $+m$ for any $h > 0$ (in particular, true for $h \to 0^+$). For $h < 0$, $\frac{P_{\mathrm{B}}}{P_{\mathrm{A}}} \to \infty$ and hence the system must be in configuration B, with magnetisation $-m$ for any $h < 0$ (in particular, true for $h \to 0^-$).

Thus, in an infinitesimal field close to zero and in the thermodynamic limit, there is a large weight assigned to states with particular magnetisation $+m$ or $-m$, for $h > 0$ and $h < 0$ respectively. This is equivalent to setting $h = 0$ but taking a *restricted ensemble*, where we exclude the configurations with the 'unpreferred' magnetisation and thus, their Boltzmann weight is identically zero. Two such probability distribution can be defined P_+ and P_- based on the magnetisation values. I haven't gone through the rigorous mathematical arguments for this, however, Goldenfeld advised to refer to [3], where this has been dealt with rigorously.

Lecture 10: Some more on Ising Model

In the previous section, we noted some properties based on the symmetry of the Ising model. Let us now continue our discussion on the mean field solution, focussing on critical exponents, that we previously saw in the phase transition in fluids.

10.1. Critical Exponents

Critical exponents are defined as the limiting *power-law* behaviour of some thermodynamic quantities near the critical point.

$$f(t) \sim t^{\lambda} \qquad \implies \qquad \lambda = \lim_{t\to 0}\frac{\ln f(t)}{\ln t}$$

The variable t is often taken to be the reduced temperature $t := \frac{\pm(T-T_c)}{T_c}$ depending on whether we want to approach from below or above T_c. The reduced temperature kinda is a measure of the distance from critical point. It is taken dimensionless so that different models can be compared.

The critical exponents for T_c^+ and T_c^- are found to be always equal, if they exist (Like for magnetisation, $m = 0$ for $T > T_c$ so no exponent can be obtained in that case and hence, magnetisation is approached only from below T_c).

From Eq. (24), the mean field magnetisation was written as $\langle\sigma\rangle = \tanh(\beta h + \beta J\gamma\langle s\rangle)$. Let us denote $m = \langle\sigma\rangle$ and $\tau = \frac{T_c}{T} = \beta J\gamma$. In terms of the reduced temperature $t = \frac{T_c - T}{T_c}$, we have $\tau = \frac{1}{1-t}$. Using $\tanh(x) \simeq x - \frac{x^3}{3}$ for $x \ll 1$, we get

$$m = \tanh(\tau m) = \tanh\left(\frac{m}{1-t}\right) = \frac{m}{(1-t)} - \frac{m^3}{3(1-t)^3} + \dots$$

Solving this, we get $m = 0$ or $m^2 = (3t)(1-t)^2$. Note that for $T > T_c$, $t < 0$ while $(1-t)^2 > 0\ \forall t$. Hence, m^2 comes out to be negative, which is not possible. Thus for $T > T_c$ we only have one solution $m = 0$. The non-zero magnetisation is thus interesting for us and let us check what is the scaling of this,

$$m^2 = (3t)(1-t)^2 = 3t + \mathcal{O}(t^2) \implies m \sim \pm(3t)^{1/2}$$

Thus we obtain the first critical exponent from the mean-field analysis,

$$m \sim \left(\frac{T_c - T}{T_c}\right)^{\beta} \qquad \text{where, } \beta = \frac{1}{2}$$

Now, let us check what happens when $h \neq 0$, along the critical isotherm $T = T_c \implies \tau = 1$. For small m and $({}^{h}/{}_{k_{\mathrm{B}}T_c}) \ll m$, the magnetisation equation becomes,

$$\begin{aligned} m = \tanh(\beta h + m) &\simeq (\beta h + m) - \frac{1}{3}(\beta h + m)^3 + \dots \\ &= (\beta h + m) - \frac{m^3}{3}\left(\frac{3\beta h}{m} + 1\right) + \dots \end{aligned}$$

From this we obtain a scaling between the field and magnetisation,

$$\beta h[1 - m^2] = \frac{m^3}{3} \implies h = \frac{m^3}{3\beta(1-m^2)} = \frac{m^3}{3\beta} + \dots \implies h \sim m^3$$

We obtained the second critical exponent from the mean-field analysis,

$$m \sim h^{1/\delta} \qquad \text{where, } \delta = 3$$

We will now deal with the third exponent, but we have to be more careful here, since this will have scaling both above and below T_c. For this, again consider the transcendental equation $m = \tanh(\beta h + \tau m)$ which, after expansion around small m and $h \ll m k_{\mathrm{B}} T$, can be written as,

$$h = \frac{m(1-\tau) + \frac{\tau^3 m^3}{3}}{\beta(1 - \tau^2 m^2)}$$

Differentiating and retaining terms only upto $\mathcal{O}(m^2)$,

$$\left(\frac{\partial h}{\partial m}\right)_T = \frac{m^2\tau^2 + (1-\tau)}{\beta\left(m^2\tau^2 - 1\right)^2}$$

The denominator cannot provide any more m^2 term and hence upto $\mathcal{O}(m^2)$, the derivative is given by,

$$\left(\frac{\partial h}{\partial m}\right)_T = \frac{1}{\beta}\left[\frac{m^2}{(1-t)^2} - \frac{t}{(1-t)}\right]$$

For $T > T_c$, $m = 0$ and for $T < T_c$, $m^2 = 3t(1-t)^2$ as found before. Substituting this above, we get

$$\begin{aligned} \chi_+ &= \left(\frac{\partial m}{\partial h}\right)_{T_c^+} = \frac{\beta(t-1)}{t} = \frac{1}{k_{\mathrm{B}}T}\left[1 - \frac{T_c}{T_c - T}\right] = \frac{1}{k_{\mathrm{B}}(T - T_c)} \\ \chi_- &= \left(\frac{\partial m}{\partial h}\right)_{T_c^-} = \frac{\beta(1-t)}{t(2-3t)} \simeq \frac{\beta(1-t)}{2t} = \frac{1}{2k_{\mathrm{B}}T}\left[\frac{T_c}{T_c - T} - 1\right] = \frac{1}{2k_{\mathrm{B}}(T_c - T)} \end{aligned}$$

We obtained the scaling of the susceptibility,

$$\begin{aligned} \chi_+ &\sim (T - T_c)^{-\gamma'} \quad \text{where, } \gamma' = 1 \\ \chi_- &\sim (T_c - T)^{-\gamma} \quad \text{where, } \gamma = 1 \end{aligned}$$

For $T \gg T_c$ then $\chi_+ \sim \frac{1}{T}$, in accordance with Curie's law. We see that $\gamma = \gamma'$ and hence, approach from both side of the critical point is the same. Also, $\frac{\chi_+}{\chi_-} = 2$ which is called the *amplitude ratio*.

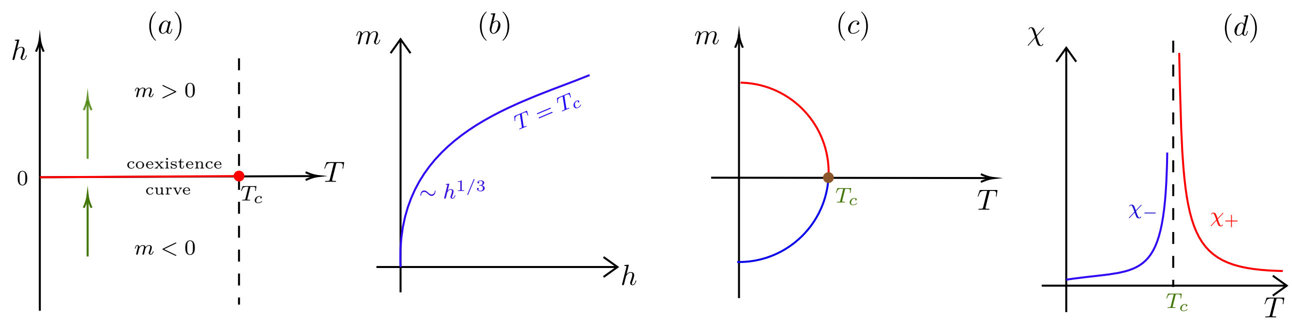


Figure 14: Scaling of various quantities in Ising model

Fig. 14 (a) shows the Ising model phase diagram. For low temperatures $T < T_c$, the system undergoes a discontinuous phase transition in which m changes abruptly as it crosses from $h < 0$ to $h > 0$, denoting a first-order transition. Shown in red is the *coexistence curve*, which ends at T_c. Beyond T_c, m changes continuously across the $h < 0$ to $h > 0$ transition.

As we vary T for a fixed h, the magnetisation varies smoothly as shown in Fig. 14 (c) until T_c, after which it becomes zero. The Ising model shows a first-order transition as a function of h but a second-order transition as a function of T.

10.2. Bragg-Williams Approximation

This is kind of an alternate mean-field description of the Ising model. Consider the Ising lattice containing N spins, with say, N_+ up spins and N_- down spins. Then the average magnetisation is given by,

$$m = \frac{N_+ - N_-}{N_+ + N_-} = \frac{2N_+ - N}{N} \implies \frac{N_+}{N} = \frac{1}{2}(m+1)$$

The entropy of the system is defined as the number of ways by which a configuration with magnetisation m can be produced. Hence, out of N spins, we choose N_+ spins and the rest are set to N_-, which gives us,

$$\Omega(N_+, N_-) = \frac{N!}{N_+!N_-!} = \frac{N!}{\left(\frac{N(1+m)}{2}\right)!\left(\frac{N(1-m)}{2}\right)!} \qquad S = k_\mathrm{B} \ln \Omega(N_+, N_-)$$

Using Sterling's approximation, $\ln N! \approx N \ln N - N$ we obtain,

$$\begin{aligned}
\ln \Omega &\approx (N \ln N - N) - \frac{N(1+m)}{2} \ln\left(\frac{N(1+m)}{2}\right) - \frac{N(1-m)}{2} \ln\left(\frac{N(1-m)}{2}\right) + N \\
&= N\left[\ln N - \frac{(1+m)}{2} \ln\left(\frac{N(1+m)}{2}\right) - \frac{(1-m)}{2} \ln\left(\frac{N(1-m)}{2}\right)\right] \\
&= N\left[\ln N - \frac{(1+m)}{2} \ln(1+m) - \frac{(1-m)}{2} \ln(1-m) - \ln N - \ln 2\right] \\
&= N\left[\ln 2 - \frac{1+m}{2} \ln(1+m) - \frac{1-m}{2} \ln(1-m)\right]
\end{aligned}$$

The mean-field energy of the Ising model is given by,

$$U = -\frac{J\gamma N m^2}{2} - Nhm$$

We can now calculate the free energy and minimise it to find the equation of state.

$$F(T, m, N) = U - TS = -\frac{J\gamma N m^2}{2} - Nhm - Nk_\mathrm{B}T\left[\ln 2 - \frac{1+m}{2} \ln(1+m) - \frac{1-m}{2} \ln(1-m)\right]$$

which implies that,

$$\frac{\partial F}{\partial m} = -JN\gamma m - Nh + \frac{Nk_{\mathrm{B}}T}{2}[1 + \ln(1+m) - 1 - \ln(1-m)] = -JN\gamma m - Nh + \frac{Nk_{\mathrm{B}}T}{2}\ln\frac{1+m}{1-m} = 0$$

$$\implies \frac{1+m}{1-m} = e^{2\beta(J\gamma m+h)} \implies m = \frac{e^{2\beta(J\gamma m+h)} - 1}{e^{2\beta(J\gamma m+h)} + 1} = \frac{e^{\beta(J\gamma m+h)} - e^{-\beta(J\gamma m+h)}}{e^{\beta(J\gamma m+h)} + e^{-\beta(J\gamma m+h)}} = \tanh(\beta(J\gamma m + h))$$

Thus we obtained the same previous self-consistent equation $m = \tanh(\beta J\gamma m + \beta h)$ from this method.

10.3. Towards the Landau free energy

Instead of writing the Ising partition function $\mathcal{Z}(T,h,N)$ as a sum over all spin configurations, we can write it as a sum over all possible magnetisations m. We can write,

$$\mathcal{Z}(T,h,N) = \sum_{m}\sum_{\{\sigma\}|m} e^{-\beta\mathcal{H}(\{\sigma\},N,h)} \approx \sum_{m}\Omega(m,N)e^{-\beta\mathcal{H}_{\mathrm{MF}}(m,N,h)}$$

where $\Omega(m,N)$ is just the *degeneracy factor*, which tells us how many states are there with magnetisation m and $\mathcal{H}_{\mathrm{MF}}$ is the mean-field Hamiltonian from Eq. (22). The above sum can be approximated by an integral from -1 to 1, taking magnetisation to be continuous:

$$\begin{aligned}
\mathcal{Z}(T,h,N) &= \int_{-1}^{1} \mathrm{d}m\ \Omega(m,N)e^{-\beta\mathcal{H}_{\mathrm{MF}}(m,N,h)} \\
&= \int_{-1}^{1} \mathrm{d}m\ e^{\ln\Omega(m,N)}e^{-\beta\mathcal{H}_{\mathrm{MF}}(m,N,h)} \\
&= \int_{-1}^{1} \mathrm{d}m\ e^{\ln\Omega(m,N)-\beta\mathcal{H}_{\mathrm{MF}}(m,N,h)} \\
&= \int_{-1}^{1} \mathrm{d}m\ e^{-\beta N\left(\frac{-J\gamma m^2}{2} - mh - \frac{k_{\mathrm{B}}T}{N}\ln\Omega(m,N)\right)}
\end{aligned}$$

The blue term feels like some kind of thermodynamic potential but it depends both on m and h. Hence this is some kind of pseudo-free energy called the *Landau free energy* denoted by $\mathcal{L}(m,h)$,

$$\mathcal{L}(m,h) = -\frac{\gamma Jm^2}{2} - mh - k_{\mathrm{B}}T\ln 2 + \frac{k_{\mathrm{B}}T}{2}[(1+m)\ln(1+m) + (1-m)\ln(1-m)]$$

From saddle-point approximation, the integral will take the value at $m = m^*$, such that $N\mathcal{L}(m^*,h)$ is minimum. Thus,

$$\left.\frac{\partial\mathcal{L}}{\partial m}\right|_{m^*} = -\gamma Jm^* - h + \frac{k_{\mathrm{B}}T}{2}[1 + \ln(1+m^*) - 1 - \ln(1-m^*)] = 0$$

From this we obtain the same transcendental equation $\tanh(\beta h + \beta J\gamma m) = m$ as before. Let us see some insights from the plot of the Landau free energy.

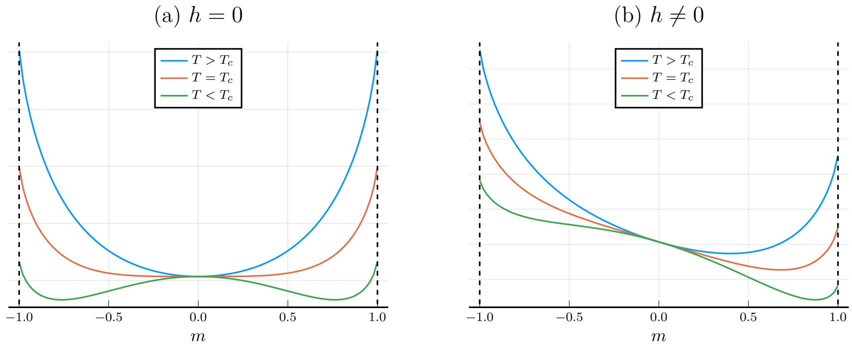


Figure 15: $\mathcal{L}$ vs. m at different temperature regimes. The curves have been appropriately shifted so that $\mathcal{L}(m = 0) = 0$

Fig. 15 (a) shows the variation of $\mathcal{L}(m)$ as m is varied between -1 and $+1$. For the zero-field case, $\mathcal{L}$ is symmetric about $m = 0$. For $T > T_c$ we have only one minima at $m = 0$ and for $T < T_c$ we have two mimima at $\pm m_0$ while $m = 0$ is a local maxima. The second derivative of the Landau free energy at $m = 0$

$$\left.\frac{\partial^2 \mathcal{L}}{\partial m^2}\right|_{m=0} = -\gamma J + \frac{k_{\mathrm{B}}T}{2}\left[\frac{1}{1+m} + \frac{1}{1-m}\right]\Bigg|_{m=0} = -J\gamma + k_{\mathrm{B}}T \implies \left.\frac{\partial^2 \mathcal{L}}{\partial m^2}\right|_{m=0,T=T_c} = 0$$

from our definition of $k_{\mathrm{B}}T_c = J\gamma$. Thus, for all values of h, $m = 0$ is an *inflection point* for $T = T_c$ where the curvature disappears, as is evident in Fig. 15.

For $T < T_c$, $\mathcal{L}$ has two degenerate minima for $h = 0$. However, when we take $h \neq 0$ then these minima becomes non-degenerate, one becomes the stable minima and one becomes the metastable one (depending on the sign of h). On increasing h, the magnetisation metastable minima keeps getting shallower and at some $h = h^*$ (called the *spinodal field*), it disappears completely as shown in Fig. 16. The condition for this is that $\frac{\partial \mathcal{L}}{\partial m} = 0$ and $\frac{\partial^2 \mathcal{L}}{\partial m^2} = 0$, which gives us

$$m - \tanh\left(\beta h + \beta J\gamma m\right) = 0 \qquad 1 - \beta J\gamma \operatorname{sech}^2\left(\beta h + \beta J\gamma m\right) = 0$$

Using $\operatorname{sech}^2(\beta h + \beta J\gamma m) = 1 - \tanh^2(\beta h + \beta J\gamma m)$ we obtain $m^* = \pm\sqrt{1 - \frac{1}{\beta J\gamma}}$. What happens is that, for $h > 0$ the metastable minima is at $m < 0$ and this is given by $m^* = -\sqrt{1 - \frac{1}{\beta J\gamma}}$ while for $h < 0$, $m^* = +\sqrt{1 - \frac{1}{\beta J\gamma}}$ is the metastable magnetisation. From these *spinodal magnetisation*, we obtain the spinodal fields by substituting this in the expression of $\beta h = \tanh^{-1}(m) - \beta J\gamma m$ according to whether $h < 0$ or $h > 0$.

$$\beta h^* = \frac{1}{2}\ln\left[\frac{1+m^*}{1-m^*}\right] - \beta J\gamma m^*$$

Hence to summarise, we have the following cases:

- $h = 0 \longrightarrow$ degenerate, equally likely minima at $\pm m_0$ for $T < T_c$.
- $h \neq 0, h < h^* \longrightarrow$ one stable and one metastable minima.
- $h \neq 0, h > h^* \longrightarrow$ only one stable minima.

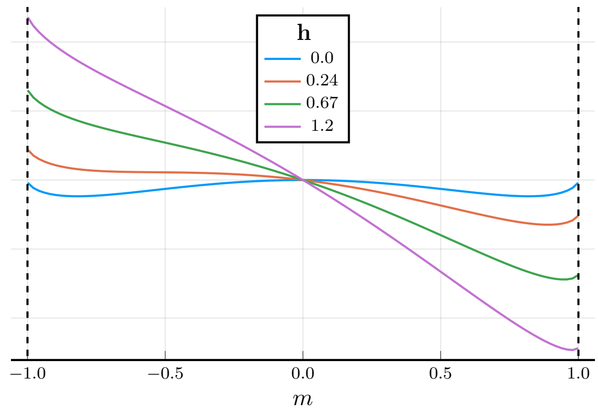


Figure 16: $\mathcal{L}$ vs. m for different values of h at $T < T_c$. On increasing h, the metastable minima at $m < 0$ disappears.

The metastable state exists only between $(-h^*, +h^*)$. At $T = T_c$ we have $\beta J\gamma = 1$ which implies $m^* = 0$ and hence $\pm h^*$ coalesce to the same point. The phase diagram for h vs. T is shown below:

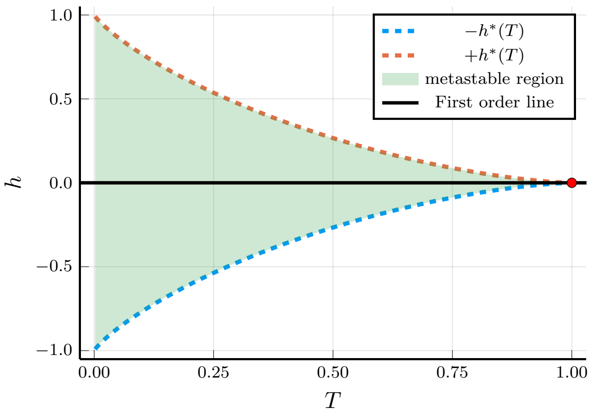


Figure 17: Phase diagram for h and T. The blue shaded region denotes the states (h, T) where the metastable regions exist. The black line denotes the first order phase coexistence curve, ending at T_c. Here T_c is taken to be 1 for the simulation.

10.4. Hysterisis

On varying the magnetic field at some fixed temperature $T < T_c$, we obtain a hysterisis in the Ising model. Suppose we start with a high negative value of $h \gg -|h^*|$. In this case, only one *global minima* is present (see Fig. 18 (a)). As we decrease the field magnitude (take the field towards zero), then at $h = -|h^*|$, the metastable minima will appear, but the system will be sitting in the global minima. At $h = 0$, the global and the metastable minima will become degenerate. On increasing h further in the positive side, the previous global minima now become the metastable minima.

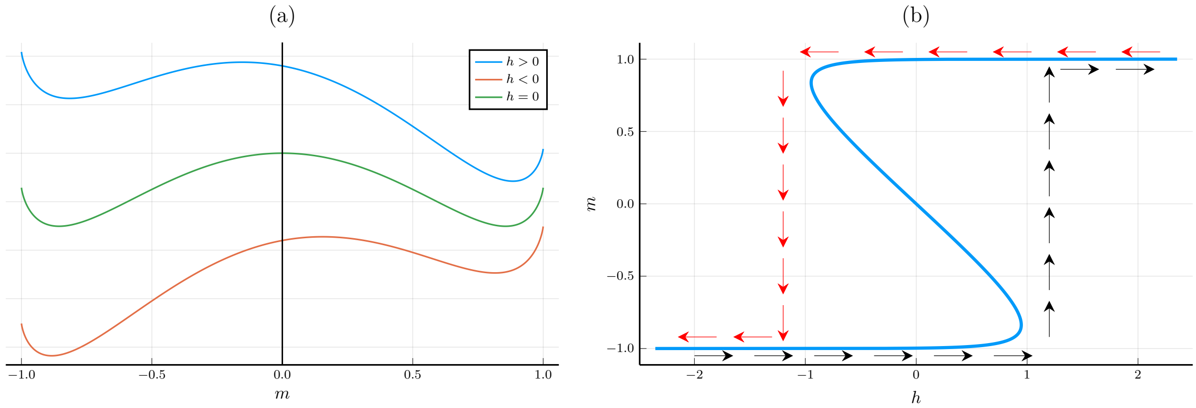


Figure 18: Hysterisis curve from mean-field theory. (a) $\mathcal{L}(m)$ vs. m in different field regimes. The global minima shifts on moving h from negative to positive. (b) Magnetisation vs. h curve, showing the hysterisis loop.

The system should technically be sitting in the global minima, however, since it was previously occupying the other minima (which has now become metastable), due to some kind of *inertia*, it still stays in the metastable state, which happens until h hits $+|h^*|$. After that, the metastable minima disappears and the system is forced to relocate to the global minima.

As shown in Fig. 18 (b), the black arrows traverse the path when we start from $h < 0$ while the red arrows show the same when starting from $h > 0$. We can see that the paths are not the same and the resulting $m(h)$ curve is not reversible in time. This is the characteristic case of *hysterisis*!

Lecture 11: Landau Theory for Phase Transition: I

We move on to a formal description of the process encountered in the previous lecture. Here, will be be doing away with the microscopic details. Although, time and again, we will be taking the Ising model as our basis to build intuition and stuff, the method can be possibly generalised to other systems. We will define the system in terms of an *order parameter* which we will take to be a scalar. However, order parameters can literally be *anything* as long as it can distinguish between two phases.

11.1. Order Parameter

Order parameters can be scalars, or in general tensors or an element from a group, it can be real or complex; there is a lot of diversity. The basic necessity is that, it should distinguish between phases. It is usually zero in the unordered phase and non-zero in the ordered phase. For example, in the *Heisenberg model* with the Hamiltonian,

$$\mathcal{H} = -J\sum_{\langle ij\rangle} \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j - \mathbf{h}\cdot\sum_i \boldsymbol{\sigma}_i \tag{29}$$

where the spins on the lattice are *vectors*, the magnetisation $\mathbf{M}$ is a vector, such that $\mathbf{M} = 0$ for $T > T_c$ and $\mathbf{M} = M\hat{\mathbf{n}}$, pointing in some arbitrary direction $\hat{\mathbf{n}}$ for $T < T_c$.

For the superfluid transition in ^{3}He, the order parameter is something called a *pairing amplitude* which is kinda a complex matrix and has to do with formation of Cooper pairs. In superconductors, the energy gap Δ is often taken to be the order parameter. In fluid systems, as seen before, we took the density difference between the phases to be the order parameter. In each of these, the number of components in the order parameter are different.

11.2. Expanding the energy

The Landau theory has to do with expanding a function $\mathcal{L}$ in powers of the *order parameter*. The function $\mathcal{L}$ must obey the symmetry of the system. With all our might we write the phenomenological expansion,

$$\mathcal{L} = \sum_{i=0}^{\infty} a_i(\{K\}, T)m^i \tag{30}$$

where $\{K\}$ denote the set of couplings in the model (for example, in the Ising model, J was our coupling constant and also h, kinda). The expansion in itself is not that helpful, unless we do something stupid with it. We expect that m should be small in and around T_c and hence we truncate the expansion upto m^4, leaving us with the quartic free energy,

$$\mathcal{L} = a_0 + a_1 m + a_2 m^2 + a_3 m^3 + a_4 m^4$$

For the time being, let us consider the $\mathbb{Z}_2$ symmetry case as in zero-field Ising model. Thus, $\mathcal{L}(m) = \mathcal{L}(-m)$ which forces the odd terms to be zero viz. $a_1 = a_3 = 0$. Moreover, we want $\mathcal{L}$ to be bounded below, since we hope to extract physical insights from minimising the free energy. If $\mathcal{L}$ has no finite minima, then we can arbitrarily make that small by taking $\eta \to \infty$ which is unphysical. Thus, $a_4 > 0$ is taken to make $\mathcal{L}$ bounded below.

All the coefficients here are dependent on temperature and the coupling. In the unordered phase, $m = 0$ and hence a_0 denotes the value of $\mathcal{L}$ in the disordered phase. Since this coefficient is independent of the order parameter, we can think of this as a *smooth background field*, containing the irrelevant degrees of freedom, upon which the phase transition stuff is superimposed. We set $a_0 = 0$, that's it! We expand the remaining coefficients with respect to the reduced temperature, $t = \frac{T-T_c}{T_c}$:

$$a_2 = a_2^{(0)} + a_2^{(1)} t + \mathcal{O}(t^2) \tag{31}$$

$$a_4 = a_4^{(0)} + a_4^{(1)} t + \mathcal{O}(t^2) \tag{32}$$

We don't need a_4 upto the linear term also, since a_4 comes with m^4, which is itself assumed to be tiny. So, we just take a_4 to be some positive constant. We now minimise $\mathcal{L}$, hoping to get some insights.

$$\frac{\partial \mathcal{L}}{\partial m} = 2a_2 m + 4a_4 m^3 = 0 \implies m = 0 \quad \text{or} \quad m = \sqrt{-\frac{a_2}{2a_4}} \tag{33}$$

We assume m to be continuous across the transition, and hence $a_2(T_c) \to 0$, which implies that $a_2^{(0)} = 0$

11.3. Continuous Transition

Lecture 12: Landau Theory for Phase Transition: II

12.1. First Order Transition

12.2. Tricritical Point

Lecture 12: Landau Theory for Phase Transition: III

Lecture 14: IDK

Lecture 15: IDK

Appendices

From the first law we have the energy conservation equation,

$$\mathrm{d}U = T\mathrm{d}S - P\mathrm{d}V + \mu \mathrm{d}N$$

and thus the internal energy is a function of S, V and N. We know that the internal energy is a homogenous function of degree 1, on physical grounds, since if we scale the system variables by λ, U is also appropriately scaled (extensitivity)

$$U(\lambda S, \lambda V, \lambda N) = \lambda U(S, V, N)$$

Now, from Euler's theorem of homogenous functions, we know that any homogenous function f of degree k can be written as,

$$kf(x_1, x_2 \dots x_n) = \sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i}$$

Using this theorem for U, we have

$$U = S\frac{\partial U}{\partial S} + V\frac{\partial U}{\partial V} + N\frac{\partial U}{\partial N}$$

Substituting the values from the first law, we obtain $U = TS - PV + \mu N$

A. Legendre Transformation

From classical mechanics, we know that the Legendre transformation is used to convert Lagrangian $L(x, \dot{x})$ to the Hamiltonian $H(x, p)$ and vice versa. Note how we switch $\dot{x}$ in L for p in H. We will call the variable that we want to switch to as the 'conjugate' of the variable we are switching. The Legendre transformation of the Lagrangian gives us the Hamiltonian,

$$H(x, p) = p\dot{x} - L$$

A function $f(x)$ with the Legendre transform $f^*(p)$ satsifies $f(x) + f^*(p) = xp$. We will use this for thermodynamic quantities also, however we will use an alternate definition using minus sign, $f(x) - f^*(p) = xp$. From the first law, we know that,

$$U = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N$$

As clearly seen, the conjugate variable pairs are (S, T), $(V, -P)$, (N, μ). We will consider the starting point as the internal energy $U(S, V, N)$ which is a function of S, V, N from the statement of the first law. However, in any experiment, keeping these as control parameters are way too hard. Think about this, controlling entropy of a system is an arduous task but controlling its conjugate variable, T, is easy-peasy.

We get four different 'potentials' by just Legendre transformation (which is basically multiplying the conjugate variables together and subtracting from the function) of the internal energy, switching between the different conjugate variables.

- **Helmholtz Free Energy**: switching $S \leftrightarrow T$

$$F(T, V, N) = U - TS \implies \mathrm{d}F = -S\mathrm{d}T - P\mathrm{d}V + \mu\mathrm{d}N$$

- **Enthalpy**: switching $V \leftrightarrow -P$

$$H(S, P, N) = U + PV \implies \mathrm{d}H = T\mathrm{d}S + V\mathrm{d}P + \mu\mathrm{d}N$$

- **Gibb's Free Energy**: switching both $S \leftrightarrow T$ and $V \leftrightarrow -P$

$$G(T, P, N) = U - TS + PV \implies \mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}P + \mu\mathrm{d}N$$

- **Grand Potential**: switching both $S \leftrightarrow T$ and $N \leftrightarrow \mu$

$$\Phi(T, V, \mu) = U - TS - \mu N = -PV \implies \mathrm{d}\Phi = -S\mathrm{d}T - P\mathrm{d}V - N\mathrm{d}\mu$$

B. 1D Solution of Ising Model

In 1D, the Ising model is given by,

$$\mathcal{H} = -J\sum_i^N \sigma_i\sigma_{i+1} - h\sum_i \sigma_i$$

where we have assumed periodic boundary conditions. The second term can be written as a symmetric sum, utilisng the translation invariance. Hence the Hamiltonian becomes,

$$\mathcal{H} = -J\sum_i^N \sigma_i\sigma_{i+1} - \frac{h}{2}\sum_i (\sigma_i + \sigma_{i+1}) \equiv \sum_i \mathcal{E}(\sigma_i, \sigma_{i+1})$$

where $\mathcal{E}(\sigma_i, \sigma_{i+1}) = -J\sigma_i\sigma_{i+1} - \frac{B}{2}(\sigma_i + \sigma_{i+1})$. Note that $\mathcal{E}(+,+) = -(J+h), \mathcal{E}(-,-) = -(J-h), \mathcal{E}(+,-) = \mathcal{E}(-,+) = J$. The contribution of each configuration is,

$$P(\{\sigma_i\}) \propto e^{-\beta\mathcal{H}(\{\sigma_i\})} = \exp\left(-\beta\sum_j \mathcal{E}(\sigma_j, \sigma_{j+1})\right) \equiv \prod_{i=1}^N \exp(-\beta\mathcal{E}(\sigma_i, \sigma_{i+1}))$$

Let us define $t_{\sigma_i\sigma_{i+1}} = \exp(-\beta\mathcal{E}(\sigma_i, \sigma_{i+1}))$ which is an entry of a matrix t called the *transfer matrix*:

$$t_{\sigma_i\sigma_j} \equiv \langle\sigma_i|t|\sigma_j\rangle$$

The transfer matrix is just a notational trick and is assumed to be an operator of the Hilbert space spanned by $|+1\rangle$ and $|-1\rangle$.

$$\begin{aligned} \langle+|t|+\rangle &= e^{\beta(J+h)} \qquad & \langle+|t|-\rangle &= e^{-\beta J} \\ \langle-|t|-\rangle &= e^{\beta(J-h)} \qquad & \langle-|t|+\rangle &= e^{-\beta J} \end{aligned}$$

The partition function can then be written as,

$$\begin{aligned} \mathcal{Z} = \sum_{\{\sigma_i\}} e^{-\beta\mathcal{H}(\{\sigma_i\})} &= \sum_{\sigma_1}\sum_{\sigma_2}\cdots\sum_{\sigma_N} t_{\sigma_1\sigma_2}t_{\sigma_2\sigma_3}\dots t_{\sigma_N\sigma_1} \\ &= \sum_{\sigma_1}\sum_{\sigma_2}\cdots\sum_{\sigma_N} \langle\sigma_1|t|\sigma_2\rangle\langle\sigma_2|t|\sigma_3\rangle\langle\sigma_3|t|\sigma_4\rangle\dots\langle\sigma_N|t|\sigma_1\rangle \end{aligned} \tag{34}$$

From the resolution of identity, $\sum_\sigma |\sigma\rangle\langle\sigma| = \mathbb{1}$ and hence all the inner spin sums vanish. We are left with,

$$\mathcal{Z} = \sum_{\sigma_1}\langle\sigma_1|t^N|\sigma_1\rangle = \mathrm{Tr}(t^N) = \lambda_+^N + \lambda_-^N = \lambda_+^N\left(1 + \left(\frac{\lambda_-}{\lambda_+}\right)^N\right)$$

where $\lambda_- < \lambda_+$ are the eigenvalues of the transfer matrix. In the limit $N \to \infty$, $\mathcal{Z} \to \lambda_+$. The eigenvalues of the transfer matrix are given as,

$$t \equiv \begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix} \implies \boxed{\lambda_\pm = e^{\beta J}\left[\cosh(\beta h) \pm \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right]}$$

In the thermodynamic limit, the free energy $F = -k_\mathrm{B}T\ln\mathcal{Z} = -Nk_\mathrm{B}T\ln\lambda_+$ and then the magnetisation is given as,

$$\begin{aligned} M = -\frac{\partial F}{\partial h} &= Nk_\mathrm{B}T\frac{\partial}{\partial h}\left[\beta J + \ln\left(\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right)\right] \\ &= Nk_\mathrm{B}T\frac{\beta\sinh(\beta h) + \frac{2\beta\sinh(\beta h)\cosh(\beta h)}{2\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}}}{\left(\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right)} \\ &= N\cancel{(k_\mathrm{B}T)}\cancel{\beta}\frac{\sinh(\beta h)\left[\sqrt{\sinh^2(\beta h) + e^{-4\beta J}} + \cosh(\beta h)\right]}{\left(\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right)\left(\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right)} \\ &= \frac{N\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}} \end{aligned}$$

The magnetisation per spin is then,

$$\boxed{m = \frac{\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}}}$$

For $T > 0$ and $h \neq 0$, m is a well-behaved function with no singularity. For $h \to 0$, we get $m \to 0$ for all $T > 0$. Thus, temperature is not sufficient to produce a finite magnetisation without an external field. For $T \to 0$, we have $m \to 1$ and hence we can say that the phase transition to ferromagnetic behaviour in the 1D Ising model is at $T_c = 0$. This is the reason why Ising said that there is no phase transition in 1D case (and incorrectly generalised it to higher dimensions too)!

References

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